

Existential Risk and Growth

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Abstract

Technology increases consumption but can create or mitigate existential risk to human civilization. Though accelerating technological development may increase the hazard rate (the risk of existential catastrophe per period) in the short run, two considerations suggest that acceleration typically decreases the risk that such a catastrophe *ever* occurs. First, acceleration decreases the time spent at each technology level. Second, given a policy option to sacrifice consumption for safety, acceleration motivates greater sacrifices by decreasing the marginal utility of consumption and increasing the value of the future. Under broad conditions, optimal policy thus produces an “existential risk Kuznets curve”, in which the hazard rate rises and then falls with the technology level and acceleration pulls forward a future in which risk is low. The negative impacts of acceleration on risk are offset only given policy failures, or direct contributions of acceleration to cumulative risk, that are sufficiently extreme.

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“If you are going through hell, keep going.”
 –Winston Churchill

1 Introduction

Technology can bring immense prosperity. Its impact on *existential risk*—the risk of human extinction, or, equivalently for decision purposes, of an equally complete and permanent loss of human welfare—strikes many as ambiguous at best.¹ Advances in vaccine technology render us less vulnerable to devastating plagues, for instance; advances in gain-of-function virology arguably make them more likely (Millett and Snyder-Beattie, 2017).

This raises the possibility of a tradeoff: concern for the long-term survival of human civilization may motivate slowing or abandoning development, at least outside of sustainability-focused domains. Sentiments along these lines from an environmentalist perspective go back at least to the Club of Rome’s 1972 report calling for a recognition of the “Limits to Growth”, and have recently reemerged with prominent calls to pause AI development (Future of Life Institute, 2023). Jones (2024) explores how to make the tradeoff between AI development and AI risk, under the assumption that the tradeoff exists.

Is this assumption justified? Would slowing technological development reduce existential risk?

To shed light on this question, we begin in Section 2 with a simple model in which the hazard rate—the probability of catastrophe per unit time—is simply a function of the technology level, and is always positive. In this setting, it is certain that an existential catastrophe will occur eventually, unless in the long run higher technology levels carry hazard rates that fall toward zero. If some existential risks are a permanent byproduct of technologically advanced civilization, civilization will avoid destroying itself technologically only if it pursues a policy of “degrowth”: self-destruction by another name.

This leaves two possibilities. If advanced technology does not eventually drive the hazard rate toward zero, then a catastrophe is inevitable, so accelerating technological development cannot increase its probability. On the

¹See e.g. Bostrom (2002), Posner (2004), Farquhar et al. (2017), Ord (2020), and Jones (2024). We will refer to the event that humanity goes extinct or suffers a similarly complete, immediate, and permanent loss of welfare as an “existential catastrophe”, or simply as a “catastrophe”.

other hand, if future technological advances will eventually drive the hazard rate to zero, so that a catastrophe is avoidable, then acceleration can lower its probability by hastening the arrival of safety.

This simple model formalizes two observations. The first, already widely appreciated, is that if we believe that the hazard rate is currently high, our only hope for a long and valuable future is the hope that we are living through a “time of perils”: a temporary period of high existential risk. This view was most famously expressed by Sagan (1997), who coined the phrase, and its implications for those especially concerned about the long-term future have been emphasized prominently by Parfit (1984), Ord (2020), and others. The second appears to be less widely appreciated: that if we are indeed in a time of perils, with the hazard rate a positive function of the technology level, then, even though technological development has increased the hazard rate so far, deceleration for the sake of long-term survival is misguided. Accelerating technological development may temporarily raise the hazard rate, but it lowers the probability that a catastrophe ever occurs.²

The model of Section 2 is not “economic”. It offers lessons about the impact on risk of quickly escaping risky states, not about optimal policy or preference maximization under constraints. It thus leaves open the possibility that, when tradeoffs between consumption and risk are navigated by a policymaker with insufficient concern for long-term survival, accelerated technological development can increase risk after all. The model of Section 2 also offers no reason to believe that future states *will* be safe. Without such a reason, if one believes that technological development has historically increased the hazard rate, the hope that this relationship will permanently reverse in the future may seem naive.

In Section 3, the heart of the paper, we therefore introduce an economic environment in which the technological frontier grows exogenously and the hazard rate is a function of the technology level and policy choices. As new potentially dangerous technologies are introduced, a planner, discounting the future at an arbitrary positive rate, decides how much potential consumption to sacrifice for the sake of lowering the hazard rate.

We find—first with a simple illustration in Sections 3.2–3.3, then more generally in Section 3.5—that the chosen policy path typically generates an

²The point is however noted informally by Bostrom (2014), p. 234, and very recently by Ord (2024).

“existential risk Kuznets curve”. That is, the hazard rate rises and then falls with time. Early in time, when the expected discounted value of the future of civilization is relatively low and the marginal utility of consumption is high, it is worthwhile to adopt risky technologies as they arrive, tolerating increases to the hazard rate for the sake of growing consumption rapidly. Later, when the expected value of the future is higher and the marginal utility of consumption has fallen, substantial risk mitigation becomes worthwhile.

The possibility of a policy response thus offers an economic justification for the view that we may indeed be living through a once-in-history time of perils. Safety is a luxury good, and technological acceleration generates a wealth effect. If the wealth effect is strong enough, then even if the hazard rate would grow indefinitely in the absence of policy, optimal policy eventually lowers drives the hazard zero. If the wealth effect is still stronger or risk mitigation still easier, then optimal policy eventually lowers the hazard quickly enough that the probability of escaping the time of perils is positive.

Because policy equates the marginal value of risk-reduction expenditures to the marginal utility of consumption, the optimal path of the hazard rate and the optimal path of consumption are closely related. Roughly speaking, the hazard rate falls to zero if the coefficient of relative risk aversion is greater than 1 (so that the marginal utility of consumption declines rapidly) and if risk mitigation is easy enough that it is not optimal for consumption to stagnate. The hazard rate falls to zero quickly enough to permit survival if, further, it is optimal for consumption to grow at at least as fast as a certain power function.

In short, details aside, policy facilitates survival.

This insight mirrors the logic of Stokey (1998) and Brock and Taylor (2005), on which environmental damages rise and then fall with economic development, and of Jones (2016, 2024), on which growth yields increases in the value of life relative to marginal consumption. Like the analysis presented here, these papers find that coefficients of relative risk aversion less than (or perhaps equal to) 1 are necessary for wealth increases to motivate large reallocations from consumption to safety. The insight above is also suggested, though briefly and informally, by Cotton-Barratt (2015), who notes that efforts to reduce existential risk may be better funded in the future because the world will be wealthier. None of these sources solve for a planner’s optimal path of a hazard rate over time, however, or characterize the conditions under which the probability of a binary event (here, existential catastrophe)

under optimal policy is less than 1.

Our model of catastrophic risk differs more significantly from those of Martin and Pindyck (2015, 2021) and Aurland-Bredesen (2019). That literature studies a society’s willingness to pay to reduce the risk of catastrophes that are, or are essentially equivalent to, proportional consumption cuts. In such a context there are no wealth effects: the fraction of consumption one is willing to sacrifice to avoid a proportional consumption cut is, by definition, independent of one’s baseline level of consumption.

In addition to facilitating survival on a *given* technology path, we find—again, first with a simple illustration in Section 3.4, then more generally in Section 3.5—that optimal policy tends to magnify the extent to which shifting from one technology path to a faster-growing technology path decreases long-term risk. As in the policy-free model, acceleration decreases the time spent in any given risky state. Under optimal policy, however, the wealthier future states pulled forward by an acceleration are systematically inclined to be safer, due to the wealth effect. Furthermore, given an increase to the future growth rate, even before actual productive capacity has yet increased, the anticipation of a more valuable future motivates more stringent safety policy in the present.

In sum, an economic analysis of the question of existential risk over time offers a possible justification for the view that we are living through a technologically-induced time of perils. At the same time, it is a justification that largely undermines a common companion of this view, namely that slowing technological development would increase the chances of civilization’s long-term survival.

This analysis might be compared with that of Baranzini and Bourguignon (1995). In a model in which growth can pose existential risk, Baranzini and Bourguignon define a growth path to be “sustainable” if it (a) minimizes the probability that an anthropogenic existential catastrophe ever occurs and (b) features non-decreasing consumption given survival. They then find conditions under which the optimal growth path, in the conventional sense of maximizing expected discounted utility, is sustainable. We do something like the reverse: we find broad conditions under which technological advances, when regulated with a view to maximizing expected discounted utility, lower the probability of an existential catastrophe.

Sections 2 and 3 explore models in which the *state* of technology at a given

time contributes to the hazard rate at the time. In Section 4, we consider the alternative possibility that risk is “transitional”, increasing in the rate of technological *development*.

We find that, in the absence of policy, the effect of acceleration on long-term transition risk is ambiguous. In particular, we observe that acceleration has no effect on long-term risk under the assumption that the “experiments” associated with a given unit of technological progress pose a degree of risk that is independent of how many experiments happen concurrently. This is the assumption employed e.g. by Jones’s (2016) “Russian roulette” model of risky technological development. If the future contains a long sequence of experiments yet to be performed, each of which carries some probability of inducing an existential catastrophe, then stagnation can lower risk by avoiding advanced experiments altogether, but a technological acceleration that only changes *when experiments occur* leaves the probability of catastrophe unchanged.

As in Section 3, introducing an optimal policy response facilitates survival due to wealth effects, potentially replacing an ever-increasing hazard rate with an existential risk Kuznets curve. Also, though the effect of acceleration on long-term transition risk remains ambiguous given policy, policy can shift the conditions under which acceleration raises, lowers, or has no effect on the probability of catastrophe. At least in the particular model of transition risk studied in Section 4, the existence of a policy response significantly widens the conditions under which acceleration lowers transition risk.

Section 5 concludes by summarizing the analysis and discussing its limitations.

2 State risk without mitigation

2.1 Model

The hazard rate

A time-varying hazard rate δ_t represents the flow probability of anthropogenic existential catastrophe.

In the (very) basic model of this section, the hazard rate is simply a

continuous function of the technology level A_t :

$$\delta_t = \delta(A_t).$$

That is, we will assume that the hazard rate at t depends entirely on the *state* of technology at t , rather than on policy choices or on the rate at which new technology is being developed.

We will assume that A_t is exogenous and strictly increasing without bound in t . In this context, further assumptions on the technology path—e.g. that it grows exponentially—can be made without loss of generality, as they simply amount to re-indexing technology levels without changing their ordering. In particular, we will assume without loss of generality that $A_{(\cdot)}$ is differentiable.

We will assume that $\delta(A) > 0$ for all A .

Survival

The probability that civilization survives to date t (starting from date 0) is given by

$$S_t \equiv e^{-\int_0^t \delta_s ds},$$

obeying the law of motion

$$\dot{S}_t = -\delta_t S_t, \quad S_0 = 1.$$

The probability that human civilization does *not* succumb to an anthropogenic existential catastrophe and, at least in expectation, enjoys a long and flourishing future³ is

$$S_\infty \equiv \lim_{t \rightarrow \infty} S_t = e^{-\int_0^\infty \delta_s ds}. \quad (1)$$

We will refer to $\{\delta_t\}_{t=0}^\infty$ as the hazard curve, to the area under the hazard curve ($\int_0^\infty \delta_t dt$) as cumulative risk, and to S_∞ as the probability of survival. Note that the probability of survival decreases in cumulative risk, and that survival is possible ($S_\infty > 0$) iff cumulative risk is finite.

³In the face of natural existential risk, this will entail eventually succumbing to a natural existential catastrophe instead. From very-long-run historical data on large-scale natural catastrophes, and the typical survival rate of other mammalian species, Snyder-Beattie et al. (2019) estimate that humanity’s natural existential hazard rate is “almost guaranteed to be less than one in 14,000” and “likely below one in 870,000” per year. Throughout this paper we ignore the possibility that technological advances may mitigate natural existential risks, but of course accounting for this possibility would only strengthen the headline results.

2.2 How does acceleration affect risk?

We will now explore how various shocks to the technology path affect the probability of survival, in this simple setting. As we will see, while the impact on risk of a temporary shock is ambiguous, a permanent level or growth effect must weakly increase the probability of survival.

2.2.1 Change of variables

Absent a negative shock severe enough to induce stagnation or recession, technology crosses every value from A_0 to ∞ exactly once. So the area under the hazard curve can be defined by integrating with respect to technology instead of time t :

$$\int_0^\infty \delta(A_t) dt = \int_{A_0}^\infty \delta(A) \left(\frac{dA}{dt} \right)^{-1} dA = \int_{A_0}^\infty \delta(A) \dot{A}_A^{-1} dA, \quad (2)$$

where, somewhat abusing notation, $\dot{A}_A$ denotes the value of $\dot{A}$ when the technology level equals the subscripted A .

2.2.2 Three ways of speeding growth

Instantaneous level effects

Consider a shock to the technology level for a short period beginning at t , so that the technology level over this period is approximately $\tilde{A}$ rather than A_t (and the subsequent technology path is unchanged). The sign of the impact of this shock on cumulative risk depends on whether $\delta(\tilde{A})$ is greater or less than $\delta(A_t)$.

More precisely, from the leftmost integral of (2), we see that the impact on cumulative risk *per unit time* of an instantaneous shock to the technology level at t , from A_t to $\tilde{A}$, equals

$$\delta(\tilde{A}) - \delta(A_t).$$

Instantaneous accelerations

Likewise, we may consider the impact on cumulative risk per unit time of an instantaneous shock to technology *growth* at t , so that the technology growth rate at t is $\tilde{\dot{A}}$ rather than $\dot{A}_t$, and subsequent technology growth rates are

unchanged. From the rightmost integral of (2), we see that the impact of this shock on cumulative risk *per unit of increase to the technology level* during the acceleration is $\delta(A_t)(\dot{\bar{A}}^{-1} - \dot{A}_t^{-1})$. Multiplying this by the new rate of technology growth per unit time, therefore, the impact on cumulative risk per unit time is

$$-\delta(A_t)(\dot{\bar{A}}/\dot{A}_t - 1).$$

Accelerations

Choose technology level $\bar{A} > A$. Since the baseline technology path increases continuously and without bound, $\bar{A} = A_T$ for some $T > t$.

Consider the effect of increasing the technology level at t from A to $\bar{A}$, and subsequently maintaining the technology path $\bar{A}_s = A_{s+(T-t)}$ ($s \geq t$). This shock to the technology path amounts to a “leap forward in time”. The impact of this shock on cumulative risk is therefore to cut a slice cut out of the hazard curve. Cumulative risk falls from (2) to

$$\int_{A_0}^{A_t} \delta(A) \dot{A}_A^{-1} dA + \int_{A_T}^{\infty} \delta(A) \dot{A}_A^{-1} dA.$$

That is, it falls by

$$\int_{A_t}^{\bar{A}} \delta(A) \dot{A}_A^{-1} dA.$$

More generally, define a temporary acceleration as an increase to $\dot{A}$ at some range of technology levels (say, from A_t to A_T). Because the exponent on $\dot{A}$ in the integral is negative, the acceleration lowers the risk endured at the given range of technology levels. A discontinuous jump in the technology level amounts to raising $\dot{A}_A$ to ∞ , and thus lowering $\dot{A}_A^{-1}$ to 0, from $A = A_t$ to A_T .

A jump in the technology level from A_t to A_T temporarily increases the hazard rate if $\delta(A_T) > \delta(A_t)$. Likewise, an acceleration to technology growth accelerates an increase to the hazard rate if $\delta(\cdot)$ is increasing around A_t . It may therefore appear to contemporaries that a given permanent level effect decreases the probability of survival.

In this context, however, that conclusion would be incorrect. If (2) is infinite, the probability of survival is zero with or without the permanent

level effect.⁴ If (2) is finite, the permanent level effect decreases cumulative risk and increases the probability of survival.

Define a permanent acceleration to be a permanent increase to $\dot{A}$ from some time t onward—or, equivalently, from some technology level A_t onward. As is plain from (2), a permanent acceleration, like a temporary acceleration, must lower cumulative risk if cumulative risk is finite on the baseline technology path.

Unlike temporary accelerations, however, permanent accelerations can render survival possible when it would otherwise have been impossible. Adding or removing a finite slice from a finite or infinite area leaves it finite or infinite respectively, but foreshortening a heavy-tailed curve with an infinite integral can yield a thin-tailed curve whose integral is finite.

To state this lesson in reverse, consider the implications of full stagnation: a permanent “negative acceleration” setting $A_s = A_t$ for all $s \geq t$. The hazard rate is then permanently positive, and so survival is impossible, even if it might have been possible at any positive technology growth rate. More concretely, consider the implications of a large negative technology shock today which returned the world to a state of ignorance about every technology developed since 1924. Perhaps the annual risk of anthropogenic existential catastrophe was much lower in 1924 than it is today, but even if so, this reset would largely doom us to relive the nuclear standoffs, emissions-intensive industrializations, and biotechnological hazards of the past. With enough replays of the past century, a catastrophe would presumably be inevitable.

Importantly, the result that stagnation must be deadly is driven by the assumption that the hazard rate is greater than zero at every technology level. Given this assumption, a positive probability of long-term survival can only be achieved by quickly driving the hazard rate toward zero, a process which requires technological innovation. Models like that of Baranzini and Bourguignon (1995), or the Jones (2016) “Russian roulette” model, produce the result that technological stagnation yields safety because they assume that, given stagnation, the hazard rate is zero, at least if it occurs at some potential levels of consumption or technological development. The implications of this alternative assumption are explored further in Section 4.

⁴ $\int_{A_t}^{A_T} \delta(A_t) dt$ is finite by the continuity of δ in A and of A in t .

3 State risk with mitigation

3.1 Motivation

If technological progress has historically increased the hazard rate, the message of the previous section is that those who wish to reduce existential risk should accelerate technological progress in the hope that the relationship between risk and technology eventually reverses. This may seem naive. Perhaps the more natural assumption is that, all else equal, technological progress will only increase the hazard rate, bringing the inevitable catastrophe sooner.

But the hazard rate presumably depends not only on technology but also on policy. If the hazard rate has increased historically, this represents a failure of policy to keep up with new risks as they have arisen. In light of the interaction between technology and policy, could existential risk be lowered by developing technology more slowly?

Trivially but importantly, accelerated technological development can indeed increase cumulative risk if the path of policy is not optimal. If policy is exogenous, for instance, cumulative risk is lower when the technology path is set so that periods of especially risky technology coincide with periods of especially stringent policy. To illustrate this, suppose that

$$\delta_t = A_t x_t,$$

where x denotes a policy variable, and that

$$x_t = (1 + t)^{-2}.$$

Then consider an acceleration from the technology path $A_t = (1 + t)^k$ to the technology path $A_t = (1 + t)^{\tilde{k}}$, where $k < 1 < \tilde{k}$. This acceleration increases cumulative risk from

$$\int_0^\infty (1 + t)^{k-2} dt \quad \text{to} \quad \int_0^\infty (1 + t)^{\tilde{k}-2} dt.$$

The former is finite, because $k - 2 < -1$. The latter is infinite, because $\tilde{k} - 2 > -1$. In this case, acceleration lowers the probability of survival to zero.

Less obvious is whether acceleration can increase cumulative risk when the policy response is optimal, within a plausible model of the feasible policy set.

One might worry that, during an interval in which more advanced technology levels carry higher hazard, a planner will adapt policy to the degree of risk, but too weakly for acceleration to lower cumulative risk on balance—perhaps in part because she cares too little about the future to be willing to sacrifice significant present consumption for the sake of safety.

To evaluate this possibility, this section introduces a policy channel through which a planner, discounting the future at an arbitrary rate, can sacrifice consumption to lower the hazard rate. As we will see, when policy is set optimally—even with respect to a high discount rate—the conclusion that acceleration lowers cumulative risk is generally not only maintained but strengthened.

As in the case where the hazard rate is an arbitrary positive function of the technology level, survival can only be achieved by pulling forward a future that asymptotically approaches perfect safety. Whereas the tech-only model of Section 2 is agnostic about whether more advanced technology will in fact carry a lower hazard rate, however, a policy response introduces a tendency for faster technological development to carry lower risk in the long run. This is for the two reasons identified by Jones (2016). First, because technology increases consumption, it decreases the utility cost of a marginal consumption sacrifice for the sake of safety. Second, when consumption is higher at a given time, the value of life, and thus the utility benefit to reducing existential risk, is higher.

Furthermore, under optimal policy, the prospect of a *future* acceleration lowers the *present* hazard rate, because when the value of the future is greater, it is worth sacrificing more today to prevent its ruin.

These dynamics are introduced in a simple model of technology and optimal policy in Sections 3.2–3.4. Their robustness is explored in the generalization of Section 3.5.

3.2 The economic environment

3.2.1 Technology

The maximum feasible level of consumption at t equals the technology level A_t . Actual consumption is A_t multiplied by a policy choice $x_t \in [0, 1]$:

$$C_t = A_t x_t. \tag{3}$$

The tradeoff at the heart of this Section 3 is that a technologically advanced civilization can risk self-destruction, but that this risk can be lowered at some cost to consumption, as represented here by a choice of x below 1. (We denote the choice variable x to remind the reader that higher choices of x come with higher existential risk.) Choices of x below 1 may constitute bans on the adoption of consumption-increasing but risky production processes, and/or allocations of resources to the production of safety-increasing goods and services, like pandemic monitoring.

The technology frontier A grows at a positive, exogenous, constant exponential rate g :

$$\dot{A}_t = A_t g, \quad g > 0, A_0 > 1. \quad (4)$$

Alternative growth paths are explored in Sections 3.5 and 4.

3.2.2 Hazard rate

A time-varying hazard rate δ_t represents the flow probability of anthropogenic existential catastrophe. The hazard rate is a function of the technology level A_t and the policy choice $x_t \in [0, 1]$, and is increasing in x_t . For now, we will assume that the elasticities of the hazard rate in A and in x are constant, so that the hazard function equals

$$\delta(A_t, x_t) = \bar{\delta} A_t^\alpha x_t^\beta, \quad \bar{\delta} > 0, \beta > \alpha > 0, \beta > 1. \quad (5)$$

We impose the three inequalities of $\beta > \alpha > 0, \beta > 1$ to satisfy three desiderata.⁵

The first is that, fixing $x_t > 0$, δ_t increase in A_t . In the context of hazard function (5), this of course requires that $\alpha > 0$. The assumption that δ_t increases in A_t is necessary if we are to concede the assessment that the development of hazardous technologies has rendered an anthropogenic existential catastrophe more likely now than it was centuries ago, and that this trend would continue in the absence of a change in policy. The proportion $1 - x$ of potential consumption sacrificed for the sake of existential safety has

⁵Hazard function (5) is closely analogous to the environmental damage function of Stokey (1998). While Stokey focuses on the implications of the damage function for the chosen path of x (or “ z ” in her notation), we will study how accelerations to the path of A affect the probability of a binary event: the occurrence of an anthropogenic existential catastrophe at any time.

only increased alongside technological development: having once been zero, it is a small but positive share today.⁶ If it had remained fixed, the hazard rate would presumably have followed a weakly higher path.

Second, the elasticity of δ_t with respect to x_t is assumed to exceed the elasticity of δ_t with respect to A_t ; i.e., $\beta > \alpha$. This is equivalent to the condition that, when technology advances, it is always feasible to lower the risk level by retaining the former consumption level, allocating all marginal productive capacity to existential safety measures. This may be seen by substituting $x_t = C_t/A_t$ (from (3)) into the hazard function (5), yielding

$$\delta_t = \bar{\delta} A_t^{\alpha-\beta} C_t^\beta.$$

Fixing C , the hazard rate falls over time iff $\beta > \alpha$. If it is (indefinitely) infeasible to lower the hazard rate while fixing consumption, as it is in this model if $\beta \leq \alpha$, then an existential catastrophe is unavoidable except through indefinite degrowth, with consumption falling to zero. This immiseration would amount to the destruction of advanced civilization by other means. In the $\beta \leq \alpha$ scenario, therefore, speeding or slowing growth can have no impact on the probability of an existential catastrophe broadly construed.

Third, fixing $A_t > 0$, δ_t is assumed to be strictly convex in x_t . This imposes $\beta > 1$. The convexity implies diminishing returns to existential risk mitigation efforts. We take this to be a reasonable assumption both from first principles and from Shulman and Thornley’s (2024) recent estimates of the cost-effectiveness of existential risk mitigation efforts, which would imply values of β over 86 (see Appendix A.1).

Generalized classes of hazard functions are discussed in Sections 3.5 and 4.

The relationship between a hazard curve and the corresponding probability of survival S_∞ is described in Section 2.1.

⁶Ord (2020, p. 313) estimates that, as of 2020, approximately \$100 million per year was spent specifically on reducing existential risk. This is likely to be a considerable underestimate of existential safety expenditures in the sense relevant here, for two reasons. First, explicit expenditures do not include foregone consumption due to regulations that slow the development or deployment of risky technologies. Second, many efforts in e.g. nuclear non-proliferation, climate change mitigation, biosecurity, and AI safety are motivated both by the desire to reduce existential risks and by the desire to reduce damages at a smaller scale. By contrast, Moynihan (2020) argues that the very concept of an anthropogenic existential catastrophe essentially did not exist 300 years ago. To the best of our understanding, there were at that time no efforts at all taken with a view to preventing one.

3.2.3 Preferences

A planner seeks to maximize

$$\int_0^\infty e^{-\rho t} S_t u(C_t) dt, \quad (6)$$

$$u(C_t) = \frac{C_t^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 1.$$

That is, flow utility $u(\cdot)$ is CRRA in consumption for some coefficient of relative risk aversion $\gamma > 1$. Flow utility is discounted at exponential rate $\rho > 0$, representing the sum of some rate of pure time preference, if any, and some rate of natural and unavoidable existential risk.⁷

The utility of death is implicitly normalized to 0 and the death-equivalent consumption level to 1. Equivalently, we are normalizing to 1 the technology level at which, when consumption is maximized, flow utility equals 0.

The planner chooses the path of x to maximize (6) subject to (3)–(5).

Like Martin and Pindyck (2015, 2021), we impose the assumption that $\gamma > 1$ throughout the paper (except in Section 3.3.4). We do this in part because this appears to the empirically relevant case, as documented by Hall (1988), Lucas (1994), Chetty (2006), and others. More importantly, however, we focus on the $\gamma > 1$ case because the results are otherwise relatively uninteresting. This is for two reasons.

First, observe that when $\gamma > 1$, flow utility is upper-bounded by $\frac{1}{\gamma-1} > 0$. Accelerating consumption growth, from a baseline of positive consumption growth, therefore yields a stream of utility benefits that eventually shrinks over time. This dynamic produces the tradeoff that motivates the paper: concern for the future may cast doubt on the value of speeding technological development, because the consumption benefits of doing so primarily accrue in the short run, whereas the costs of an existential catastrophe are everlasting. By contrast, when $\gamma \leq 1$, flow utility grows in consumption without bound, so accelerations to consumption growth and reductions in existential risk can have comparable long-term benefits.

⁷It is not necessary to microfound the preferences above, but one valid interpretation would be that the population is fixed and (6) is the expected utility of a representative household. Another would be that population grows exponentially at rate $n < \rho$, that the rate of pure time preference and exogenous risk is in fact $\rho + n$, and that the planner employs the total utilitarian social welfare function.

Second and relatedly, when $\gamma \leq 1$, the marginal utility of consumption does not decline quickly enough (relative to the rising value of civilization) to motivate rapid increases in consumption sacrifices for the sake of safety. As a result, the probability of long-term survival is always zero on the planner's chosen path, and accelerations or decelerations to technological development have no impact on the probability. This is detailed in Section 3.3.4.

3.3 The existential risk Kuznets curve

3.3.1 Optimality

Summarizing the environment of Section 3.2, the planner's problem is to choose $\{x_t\}_{t=0}^{\infty}$ to maximize

$$\int_0^{\infty} e^{-\rho t} S_t u(C_t) dt, \quad (7)$$

$$u(C_t) \equiv \frac{C_t^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 1 \quad (8)$$

subject to

$$\begin{aligned} A_0 &> 1, \\ \dot{A}_t &= gA_t \quad (g > 0), \\ C_t &= A_t x_t, \\ S_0 &= 1, \\ \dot{S}_t &= -\delta_t S_t, \\ \delta_t &= \bar{\delta} A_t^{\alpha} x_t^{\beta} \quad (\bar{\delta} > 0, \beta > \alpha > 0, \beta > 1). \end{aligned} \quad (9)$$

This section finds the path of the hazard rate in the planner-optimal solution, observing that it rises and then falls with time. In the subsequent section we will explore what this implies for the impact of speeding growth on the probability of an anthropogenic existential catastrophe. From now on, we will typically refer to such an event simply as a “catastrophe”.

The planner faces one choice variable, x_t , and one state variable, S_t . Her (expected) flow payoff at t is $S_t u(C_t)$. Her problem can be represented by

the following current-value Lagrangian:

$$\begin{aligned}\mathcal{L}_t &= S_t u(C_t) + v_t \dot{S}_t + \mu_t(1 - x_t) \\ &= S_t \frac{(A_t x_t)^{1-\gamma} - 1}{1-\gamma} - v_t \bar{\delta} A_t^\alpha x_t^\beta S_t + \mu_t(1 - x_t).\end{aligned}\quad (10)$$

μ_t is the Lagrange multiplier on the policy choice, positive iff the $x_t \leq 1$ constraint binds.

$$v_t = \int_t^\infty e^{-\rho(s-t)} \frac{S_s}{S_t} u(C_s) ds \quad (11)$$

is the costate variable on survival: the expected value of the future of civilization at t , conditional on survival to t .⁸

On an optimal path, the first-order condition on (10) with respect to the choice variable x_t is satisfied. Differentiating (10) with respect to x_t , we have

$$S_t A_t^{1-\gamma} x_t^{-\gamma} - \bar{\delta} A_t^\alpha \beta x_t^{\beta-1} v_t S_t \geq 0, \quad (12)$$

with inequality iff the left-hand side is positive at $x_t = 1$, in which case $x_t = 1$ is optimal.⁹ Thus,

- As long as (12) is nonnegative at $x_t = 1$, the optimal choice of $x_t \in [0, 1]$ equals 1. Even the first marginal sacrifices of consumption would come with greater flow costs than expected benefits.
- When (12) is negative at $x_t = 1$, the optimal choice of x_t is interior. It sets (12) equal to zero, maintaining the condition that the marginal cost to flow utility of lowering consumption equals the expected benefit via risk reduction.¹⁰

In fact there is a unique¹¹ optimal path, characterized by first-order condition (12), a first-order condition corresponding to the state variable S_t , and

⁸The fact that the costate variable on survival must equal (11) can be seen immediately by reflecting on the fact that, in effect, the value of saving the world must equal the value of the world. It is also derived formally in Appendix B.1.

⁹The second derivative with respect to x_t is negative by the assumption that $\beta > 1$.

¹⁰We can ignore the possibility that the optimal choice of x_t equals 0 because such a choice yields infinite flow disutility.

¹¹Under the restriction of piecewise continuity. If path x is optimal, measure-zero deviations from x are of course also optimal.

identity (11). This is shown in Appendix B.1 for the strictly more general environments of Section 3.5 and 4. For now, our discussion will rely only on the observations that on any optimal path, (12) is satisfied, and that on any feasible path, v_t is upper-bounded by

$$\bar{v} \equiv \frac{1}{\rho(\gamma - 1)}. \quad (13)$$

3.3.2 Initial risk increases

The condition that (12) is nonnegative at $x_t = 1$ is equivalent to the condition that

$$A_t^{-(\alpha+\gamma-1)} \geq \bar{\delta}\beta v_t. \quad (14)$$

The continuation value of civilization at t given survival to t , v_t , always strictly rises over time. This follows from the fact that, given the optimal paths $\{C_s\}_{s \geq t}$ and $\{\delta_s\}_{s \geq t}$ achievable at a given initial technology level A_t , a higher initial technology level allows for a path with an equal hazard rate but more consumption at each future period, by the assumption that $\beta > \alpha$. A higher initial technology level always enables the planner to implement a preferred future.

Therefore, early in time, when A_t is low, inequality (14) is satisfied. The optimal policy choice is $x = 1$, and the hazard rate rises with A at rate

$$g_{\delta t} = \alpha g.$$

We will assume that time 0 is defined to be early enough in time that inequality (14) is satisfied strictly at $t = 0$.

3.3.3 Eventual risk declines and survival

As the left-hand side of (14) falls exponentially with A_t and the right-hand side rises, there is a unique time t^* at which (14) holds with equality. After t^* , the optimal choice of x_t is interior and sets (12) equal to zero.

Setting (12) equal to zero and rearranging, we have the optimal choice of x_t after t^* , and thus the optimal choice of x_t in general:

$$x_t = \begin{cases} 1, & t \leq t^*; \\ (\bar{\delta}\beta A_t^{\alpha+\gamma-1} v_t)^{-\frac{1}{\beta+\gamma-1}}, & t > t^*. \end{cases} \quad (15)$$

Taking the growth rate of each side, we can find the growth rate of the policy choice variable after t^* :

$$g_{xt} = -\frac{\alpha + \gamma - 1}{\beta + \gamma - 1}g - \frac{1}{\beta + \gamma - 1}g_{vt}, \quad (16)$$

where, given a time-dependent variable y , $g_{yt} \equiv \dot{y}_t/y_t$ denotes its proportional growth rate at t . The hazard rate in turn grows as

$$\begin{aligned} g_{\delta t} &= \alpha g + \beta g_{xt} \\ &= -\frac{(\beta - \alpha)(\gamma - 1)}{\beta + \gamma - 1}g - \frac{\beta}{\beta + \gamma - 1}g_{vt}. \end{aligned} \quad (17)$$

Because $\beta > \alpha$ and $\gamma > 1$, (17) is negative.

Furthermore, though g_{vt} is always positive, $g_{vt} \rightarrow 0$. This roughly follows from the fact that the expected value of the future v_t is bounded above by $\bar{v}$.¹² This gives us the asymptotic long-run negative growth rates g_x and g_δ .

Finally, since $C_t = A_t x_t$, we have

$$\begin{aligned} g_{Ct} &= g + g_{xt} \\ &= \frac{\beta - \alpha}{\beta + \gamma - 1}g - \frac{1}{\beta + \gamma - 1}g_{vt}. \end{aligned}$$

Because $\beta > \alpha$, long-run consumption growth is positive: though x declines to 0, A grows more quickly than x declines. Indeed, the growth of consumption is key to the growth in sacrifices for the sake of safety. In the face of decreasing marginal utility to *consumption* and decreasing marginal returns to *safety effort*, potential consumption increases are split between the former and latter so that the marginal value of each remains equal.

To summarize:

Proposition 1. *The existential risk Kuznets curve*

On the planner-optimal path defined by (7)–(9), there exists a time t^ such that for $t \leq t^*$,*

$$\begin{aligned} x_t &= 1, \\ g_{Ct} &= g > 0, \\ g_{\delta t} &= \alpha g > 0 \end{aligned}$$

¹²The $g_{vt} \rightarrow 0$ limit is shown formally in Appendix B.2.

and for $t > t^*$,

$$\lim_{t \rightarrow \infty} g_{xt} = -\frac{\alpha + \gamma - 1}{\beta + \gamma - 1}g < 0, \quad (18)$$

$$\lim_{t \rightarrow \infty} g_{Ct} = \frac{\beta - \alpha}{\beta + \gamma - 1}g > 0,$$

$$\lim_{t \rightarrow \infty} g_{\delta t} = -\frac{(\beta - \alpha)(\gamma - 1)}{\beta + \gamma - 1}g < 0 \quad (19)$$

with all three limits approached from below.

Corollary 1.1. *Survival*

On the planner-optimal path defined by (7)–(9), $S_\infty > 0$.

Proof. The result follows from (19) and the definition of S_∞ . Because δ_t ultimately falls exponentially, $\int_0^\infty \delta_t dt < \infty$, so $S_\infty \equiv e^{-\int_0^\infty \delta_t dt} > 0$.

Note that $\delta_t \rightarrow 0$ is insufficient for survival. If δ_t fell to 0 too slowly, the integral would diverge, and we would have $S_\infty = 0$. $\square$

3.3.4 No survival with $\gamma \leq 1$

As noted in Section 3.2.3, one reason for focusing on the $\gamma > 1$ case is that, when the marginal utility of consumption declines too slowly, a rapid shift from consumption to safety effort is not implemented, and the probability of long-term survival is always zero.

This result recalls the “Russian roulette” model of Jones (2016). There, it is found that if $\gamma < 1$, a planner will choose to let risky technological development continue indefinitely, despite periodically posing life-threatening catastrophes, whereas if $\gamma \geq 1$, the planner eventually halts growth and thus implicitly bounds the probability that a life-threatening technologically induced catastrophe ever occurs. In that model, however, risk is posed by the development, rather than the existence, of advanced technologies. It is thus more closely analogous to (indeed, essentially a special case of) our “transition risk” model of Section 4, and is discussed further there.

The result also recalls, and sharpens, Jones’s (2024) observation about the importance of the coefficient of relative risk aversion for the willingness to avoid existential risk. Jones observes, in a single-period setting, that when γ is low, a planner should be willing to tolerate a high risk of existential catastrophe in exchange for a spurt to consumption growth. In a single-period

setting, the tolerated risk is continuous in γ ; no discontinuity is observed at $\gamma = 1$. In the dynamic setting studied here, however, a planner effectively chooses how much risk to tolerate period after period. When $\gamma \leq 1$, enough risk is tolerated each period that an eventual catastrophe is guaranteed.

Proposition 2. Policy choice and risk with $\gamma \leq 1$

Suppose a planner faces problem (7)–(9), but with utility function (8) replaced by

$$u(C_t) = \begin{cases} \log(C_t), & \gamma = 1; \\ \frac{C_t^{1-\gamma} - 1}{1-\gamma}, & \gamma < 1 \end{cases} \quad (20)$$

for some $\gamma \leq 1$, and

$$\rho > \underline{\rho} \equiv \frac{(\beta - \alpha)(1 - \gamma)}{\beta} g \quad (21)$$

to ensure the existence of an optimal policy.

Then there exists a time t^* such that for $t \leq t^*$,

$$\begin{aligned} x_t &= 1, \\ g_{Ct} &= g > 0, \\ g_{\delta t} &= \alpha g > 0 \end{aligned}$$

and for $t > t^*$,

$$\lim_{t \rightarrow \infty} g_{xt} = -\frac{\alpha}{\beta} g < 0, \quad (22)$$

$$\lim_{t \rightarrow \infty} g_{Ct} = \frac{\beta - \alpha}{\beta} g > 0,$$

$$\lim_{t \rightarrow \infty} \delta_t t = \frac{\rho}{(\beta - \alpha)g} > 0, \quad \gamma = 1; \quad (23)$$

$$\delta^* \equiv \lim_{t \rightarrow \infty} \delta_t = \frac{(\rho - \underline{\rho})(1 - \gamma)}{\beta + \gamma - 1} > 0, \quad \gamma < 1. \quad (24)$$

Proof. See Appendix B.2. □

Corollary 2.1. No survival with $\gamma \leq 1$

On the planner-optimal path defined by (7)–(9), with utility function (8) replaced by (20), $S_\infty = 0$.

Proof. The result follows from (23)–(24) and the definition of S_∞ . When δ_t is asymptotically constant or proportional to $1/t$, $\int_0^\infty \delta_t dt = \infty$, so $S_\infty \equiv e^{-\int_0^\infty \delta_t dt} = 0$. $\square$

The case in which δ_t declines proportionally to $1/t$, obtained by $\gamma = 1$, is the edge case in which the expected length of time until a catastrophe is infinite even though the probability of catastrophe is 1.

Though a catastrophe is here inevitable on the chosen path, it can be seen from (24) that faster technology growth g lowers the asymptotic hazard rate δ^* when $\gamma < 1$. This is essentially because, when $\gamma < 1$, consumption and thus flow utility grow at a higher exponential rate in the long run when g is higher, so the effect of raising g is similar to the effect of decreasing the discount rate ρ .

There is not a general result that increases to g always increase the “life expectancy of civilization” when $\gamma < 1$, however. This is discussed briefly at the end of Section 2.2.2.

Understanding the path of policy choice and risk is somewhat more complex when $\gamma \leq 1$ than when $\gamma > 1$, because we do not have the result that v_t is asymptotically constant, but a sketch is as follows.

As in the $\gamma > 1$ setting of Proposition 1, early in time inequality (14) holds and it is optimal to set $x_t = 1$. Likewise, later in time, optimality requires setting $x_t < 1$ so as to maintain

$$\begin{aligned} A_t u'(C_t) &= \frac{\partial \delta}{\partial x} \cdot v_t \\ \implies A_t x_t C_t^{-\gamma} &= \bar{\delta} A_t^\alpha \beta x_t^\beta v_t \\ \implies \delta_t &= \frac{C_t^{1-\gamma}}{\beta v_t}. \end{aligned} \tag{25}$$

Observe from (11) that v_t grows roughly with flow utility $u(C_t)$. Flow utility, for large C_t , then grows approximately like $C_t^{1-\gamma}$ when $\gamma < 1$. The result is that, though consumption grows exponentially in the long run for any value of γ , δ is asymptotically constant when $\gamma < 1$.

Intuitively, for the policy path to be optimal, it must maintain

- a) the flow utility to proportionally increasing consumption, $C_t \cdot C_t^{-\gamma}$
- =

- b) the damage done via proportionally raising the hazard rate,
 which equals the hazard rate $\times$ the value of civilization.

When the value of civilization also grows like $C_t^{1-\gamma}$, as it does when $\gamma < 1$, the hazard rate must be constant for (a) and (b) to grow at the same rate. When $\gamma > 1$, the value of civilization is asymptotically constant, so the hazard rate falls like $C_t^{1-\gamma}$.

When $\gamma = 1$, given that consumption grows exponentially, $\log(C_t)$ and thus v_t grow linearly. The hazard rate then falls proportionally to $1/t$.

To focus on the scenarios in which accelerations to technological development can affect the probability of survival at all, and for simplicity, we will maintain the $\gamma > 1$ assumption throughout the remainder of the paper.

3.3.5 Simulation

The paths of policy choice and the hazard rate are simulated below, for the following parameter values:

ρ	0.02	δ	0.00012
γ	1.5	α	1
g	0.02	β	2
A_0	2		

Table 1: Simulation parameters for Figure 1

The values of ρ , γ , and g have been chosen as central estimates from the macroeconomics literature. $A_0 = 2$ is chosen so that the value of a statistical life-year at $t = 75$ is four times consumption per capita, roughly matching estimates from Klenow et al. (2023).¹³ That is, the first year of the simulation might be taken to denote 1949, the year at which a nuclear war between superpowers first became possible, and the 75th year might be taken to denote the time of writing. $\bar{\delta}$, α , and β are chosen so that the hazard rate today

¹³They estimate that this ratio was approximately 5 in the United States in 2019. The figure must be adjusted upward in light of economic growth since 2019, but downward insofar as the model is intended to describe the path of optimal policy across all countries advanced enough to be deploying existentially hazardous technology, including many which are poorer than the United States.

is approximately 0.1%, matching Stern's (2007) oft-cited figure; so that the hazard rate begins to fall at approximately $t = 100$; and so that the growth rate and then the decay rate of the hazard rate are non-negligible, for clarity in illustration.

The probability of survival S_∞ under these parameters, from $t = 75$ onward, is approximately 65%.

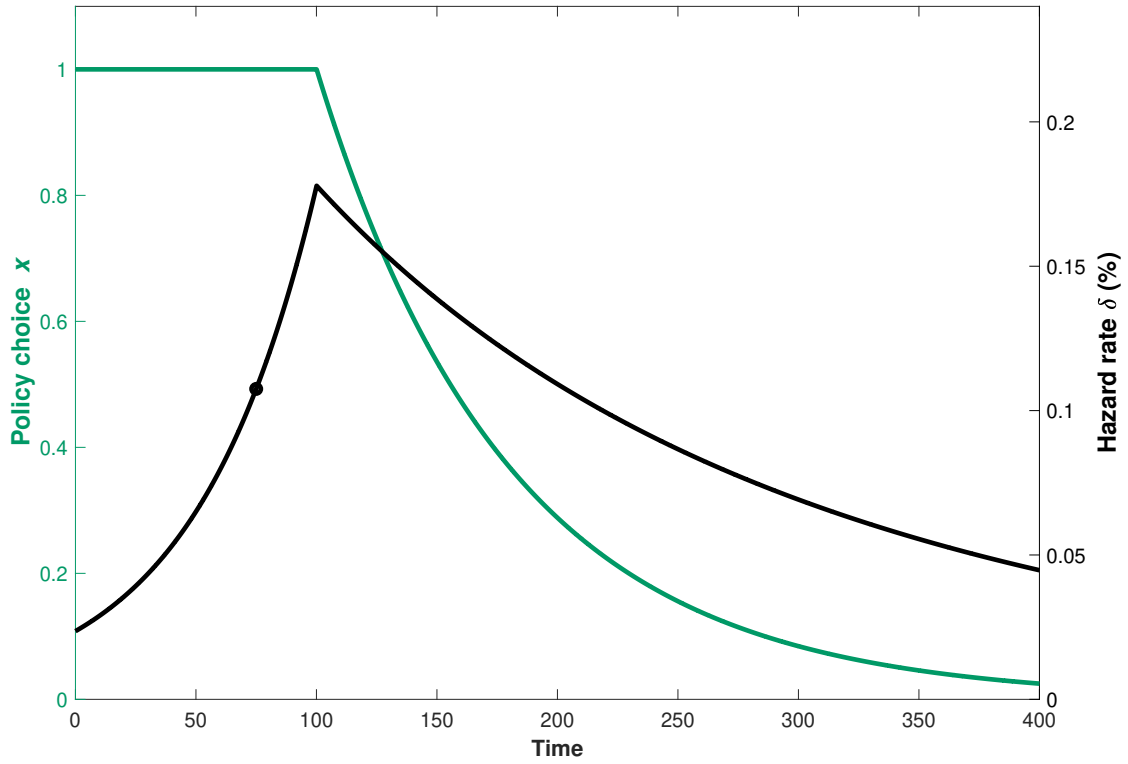


Figure 1: Evolution of the policy choice and the hazard rate along the optimal path

Calculations and code for replicating the simulation and corresponding probability of survival may be found in Appendix C.

As Figure 1 illustrates, one potentially unappealing feature of the baseline model is that it implies that, on the optimal path, the hazard rate only rises

during the regime in which no sacrifices whatsoever are made for existential safety. In this respect it closely resembles Stokey’s (1998) “environmental Kuznets curve”, which also features damages which rise exponentially with economic growth and then fall sharply past the point at which it first becomes optimal to take action. As discussed in Section 3.2.2, this pattern may be at odds with the experience of the last century, during which the hazard rate has arguably risen alongside existential risk mitigation efforts.

As in Stokey (1998), this dynamic is essentially driven by the lack of a lower Inada condition on $1 - x$. If marginal “safety expenditures” lower the hazard rate infinitely per unit spent at $x = 1$, then as long as $v_t > 0$ it is optimal to set $x_t < 1$, even if early in time the hazard rate is allowed to rise. Rising δ can thus be found alongside falling x by tweaking the behavior of the hazard function around $x = 1$. Such tweaks do not affect the long-run behavior of the policy choice or the hazard rate as given by (18)–(19), which are determined by the behavior of the hazard function around $x = 0$. This is discussed further in Appendix A.3.1.

3.4 Acceleration and risk

The analysis of the previous section lets us determine how various shocks to the technology growth path affect the probability of survival in the planner’s solution. As in the tech-only model of Section 2, the impact on risk of a temporary shock is ambiguous, but the impact of a permanent level or growth effect is always to lower risk.

The impact of a shock to growth on the probability of survival is explored for a more general class of hazard functions in Section 3.5.4, and the results are stated formally there in Proposition 6. Here we will illustrate the dynamics with more discussion using hazard function (5) in particular.

3.4.1 Change of variables

As in the tech-only model of Section 2, absent a negative shock severe enough to induce stagnation or recession, A crosses every value from A_0 to ∞ exactly once, so the area under the hazard curve can be defined by integrating with respect to A instead of t . We will let X denote cumulative risk given that

the technology path is $A_{(\cdot)}$ and the policy path x is optimal given $A_{(\cdot)}$:

$$\begin{aligned} X &\equiv \int_0^\infty \bar{\delta} A_t^\alpha x_t^\beta dt = \int_{A_0}^\infty \bar{\delta} A^\alpha x_A^\beta \left(\frac{dA}{dt}\right)^{-1} dA \\ &= \int_{A_0}^\infty \bar{\delta} A^\alpha x_A^\beta \dot{A}_A^{-1} dA, \end{aligned} \quad (26)$$

where we will again abuse notation somewhat by letting x_A and $\dot{A}_A$ denote, respectively, the optimal value of x (given technology path $A_{(\cdot)}$) and the value of $\dot{A}$ when the technology level equals the subscripted A .

We will define v_A and δ_A likewise. Note that $\delta_A \equiv \bar{\delta} A^\alpha x_A^\beta$, without dividing this expression by $\dot{A}_A$. That is, it is still a hazard rate: it represents the probability of catastrophe per unit time at technology level A , not the probability of catastrophe per unit of technological development.

3.4.2 Three ways of speeding growth

Instantaneous level effects

The effect per unit time of a positive shock to the technology level A_t , letting policy adjust instantaneously, depends on whether the shock occurs before or after the regime-change time t^* .

At $t < t^*$, temporarily multiplying the technology level by $m > 1$ has no impact on the optimal choice of x .¹⁴ The hazard rate thus rises. The future hazard rate is unaffected, so cumulative risk increases by

$$\delta_t(m^\alpha - 1) > 0$$

per unit of time that the technology level is raised.

At $t \geq t^*$, temporarily multiplying the technology level by $m > 1$ multiplies the policy variable by $m^{-\frac{\alpha+\gamma-1}{\beta+\gamma-1}}$, by (15). In combination, the positive shock to technology and the negative impact on the policy variable multiply the hazard rate by $m^{\alpha-\beta\frac{\alpha+\gamma-1}{\beta+\gamma-1}} = m^{-\frac{(\beta-\alpha)(\gamma-1)}{\beta+\gamma-1}} < 1$. This resulting change in cumulative risk is

$$\delta_t\left(m^{-\frac{(\beta-\alpha)(\gamma-1)}{\beta+\gamma-1}} - 1\right) < 0$$

per unit of time that the technology level is raised.

¹⁴Unless m is large enough to reverse inequality (14), a case we will ignore for simplicity.

Instantaneous accelerations

Multiplying the technology growth rate at t by $m > 1$ lowers cumulative risk (per unit of time that the shock lasts) regardless of t . It does so only because the shock decreases the time spent at technology levels around A_t . The shock has no impact on the policy associated with any technology level.

As in the tech-only model, therefore, we see that the impact of this shock on cumulative risk *per unit of increase to the technology level* during the acceleration is

$$\delta_t((m\dot{A}_t)^{-1} - \dot{A}_t^{-1}) < 0.$$

So the impact on cumulative risk *per unit of time* that the acceleration lasts is the above multiplied by the new technology growth rate $m\dot{A}_t$:

$$\delta_t(1 - m) < 0.$$

Accelerations

Consider the effect of raising the technology level at t from A_t to $\bar{A} > A_t$ and subsequently maintaining exponential technology growth. Observe that the initial value of technology alone determines the optimal subsequent path of policy. As in the tech-only model, therefore, this technology shock amounts to a “leap forward in time”. The resulting impact on cumulative risk is

$$- \int_{A_t}^{\bar{A}} \delta_A \dot{A}_A^{-1} dA.$$

More generally, define an acceleration at t as an increase in the technology growth rate from $\dot{A}_A$ to $\dot{\tilde{A}}_A \in (\dot{A}_A, \infty]$ for $A \in [A_t, \bar{A})$, $\bar{A} \in (A_t, \infty]$. Let $\tilde{A}_{(\cdot)}$ denote the new technology path.¹⁵ An acceleration can lower the risk endured at the given range of technology levels for two reasons.

1. As in the tech-only model, increasing the technology growth rate at A always lowers cumulative risk directly because the exponent on $\dot{A}_A$ in integral (26) is negative: $\dot{\tilde{A}}_A^{-1} < \dot{A}_A^{-1}$.

¹⁵We will let $\tilde{\delta}_A$ denote the hazard rate at A given technology path $\tilde{A}_{(\cdot)}$ for $A < A_t$ and the hazard rate at A given technology path $\tilde{A}_{(\cdot)}$ for $A \geq A_t$. We will define $\tilde{\delta}_t$, $\tilde{x}_A$, $\tilde{X}$, etc. likewise.

2. Going beyond the tech-only model, given $A \in [A_t, \bar{A})$, the value of the future at A is higher under the path with faster future technology growth: $\tilde{v}_A > v_A$. By (15), this motivates weakly more stringent policy $\tilde{x}_A \leq x_A$ and thus a weakly lower hazard rate $\tilde{\delta}_A \leq \delta_A$. If $\dot{\tilde{A}}_A^{-1} > 0$, a lower hazard rate at A lowers cumulative risk.

Via the first channel alone, the change in cumulative risk achieved by an acceleration is the integral, across technology levels, of the risk reductions achieved by instantaneous accelerations at each technology level:

$$\int_{A_t}^{\bar{A}} \delta_A (\dot{\tilde{A}}_A^{-1} - \dot{A}_A^{-1}) dA < 0.$$

Given a policy impact, the cumulative risk reduction achieved by an acceleration is greater.

Since we assume here that the baseline technology path is exponential, a permanent level effect is equivalent to a temporary acceleration (one for which $\bar{A} < \infty$). A growth effect is a special case of a permanent acceleration (one for which $\bar{A} = \infty$).

The effects of a sharp permanent level effect are illustrated in Figure 2. The parameter values used to illustrate the baseline path are the same as those used to simulate Figure 1. The level effect takes place “today”, at $t = 75$, and multiplies A by $e^{0.2} \approx 1.22$, so that at $g = 0.02$, it amounts to a 10-year leap forward.

Recall from Section 3.3.5 that the probability of survival (from $t = 75$ onward) on the baseline path is approximately 65%. The proportional increase in the probability of survival can be found analytically. Cumulative risk X declines by precisely the area under the baseline hazard curve from $t = 75$ to 85; and since $\delta_{75} = 0.1\%$, $g = 0.02$, and $\alpha = 1$, this difference equals

$$\Delta X = -0.001 \int_0^{10} e^{0.02t} dt = -0.05(e^{0.2} - 1).$$

$S_\infty = e^{-X}$ is then multiplied by

$$e^{-\Delta X} \approx 1.011,$$

so that in absolute terms S_∞ rises by approximately $0.65 \cdot 0.011 \approx 0.7\%$.

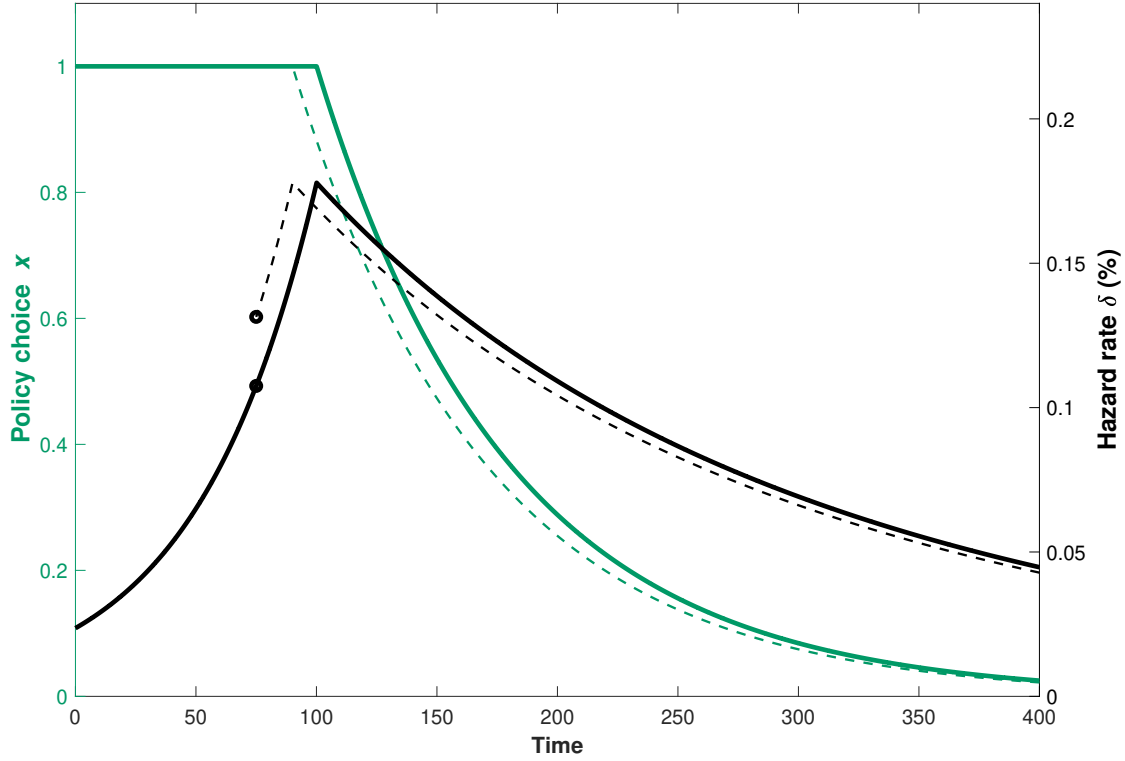


Figure 2: A level effect to growth shrinks cumulative risk

Calculations and code for replicating the simulation may be found in Appendix C.

As noted in Section 2.2, a permanent negative acceleration can render survival impossible when it would otherwise have been possible, e.g. when it induces complete stagnation.¹⁶

¹⁶Complete stagnation is not necessary to foreclose survival. Consider a negative growth effect after which the technology level grows not exponentially but power-functionally, so that $\tilde{A}_t = t^k$ for some $k > 0$. The exponential growth rate of $\tilde{A}$, which we will denote $\tilde{g}$, is then not constant at g but time-varying, with $\tilde{g}_t = k/t$. By (17) and since $\tilde{g}_v \rightarrow 0$, it then follows that $\tilde{\delta}_t$ falls to 0 like $t^{-\frac{(\alpha-\beta)(\gamma-1)}{\beta+\gamma-1}k}$. Since cumulative risk is finite for $\delta_t \propto t^{-\kappa}$ iff $\kappa > 1$, the probability of survival is positive given $\tilde{A}_t = t^k$ iff $k > \frac{\beta+\gamma-1}{(\alpha-\beta)(\gamma-1)}$.

Acceleration can lower life expectancy

As we have seen, accelerations always increase the probability of survival under the assumptions of Section 3.2: in particular, under the assumption that $\gamma > 1$. One might thus expect that in the $\gamma < 1$ case of Section 3.3.4, in which catastrophe is inevitable and civilizational “life expectancy” is finite, accelerations increase this life expectancy. However, this does not necessarily hold.

To see this most simply, observe that at any value of γ , stagnation at a low technology level A yields a permanent hazard rate of $\bar{\delta}A^\alpha$. This may be arbitrarily low, so the expected duration until catastrophe $1/(\bar{\delta}A^\alpha)$ may be arbitrarily high. When $\gamma < 1$, an acceleration can quickly yield a hazard rate that permanently approximates δ^* . The growth effect can thus lower civilizational life expectancy to approximately $1/\delta^*$.

3.4.3 Patience vs. growth

The key mechanism at work throughout this section is that as consumption grows, the planner’s willingness to sacrifice consumption for safety rises. By contrast, those concerned about the security of the long-term future often prioritize moral persuasion, appealing to ethical arguments for a low rate of pure time preference. Consider e.g. the Stern–Nordhaus debate (and the long debate since) over the social discount rate to use in the context of climate policy, or the arguments for concern for the future put forward by philosophers such as Parfit (1984), Cowen and Parfit (1992), and more recently Ord (2020) and MacAskill (2022). How do these two mechanisms—a permanent level effect to A vs. a permanent reduction to the rate of pure time preference ρ —compare in terms of increasing the probability of survival?

We will see that, early in time, decreases to ρ are arbitrarily more impactful than increases to A . Late in time, however, the impacts of the two interventions are comparable.

A sharp and permanent level effect at t , whereby A is multiplied by m slightly greater than 1, amounts to a leap forward of approximately m/g years. This decreases cumulative risk by approximately $\delta_t m/g$.

Before the regime-change time t^* , therefore, the impact of a level effect on cumulative risk rises exponentially with δ_t over time. Early in time, when A_t and δ_t are arbitrarily low, the impact of the level effect on cumulative risk

is arbitrarily low. The impact of a decrease to ρ on cumulative risk, on the other hand, does not change over time before t^* . A decrease to ρ does not affect the hazard rate immediately, but decreases it in the future by pulling forward the regime-change time and changing the path of the hazard rate afterward. These impacts do not depend on *when* (before t^*) ρ is lowered.

By contrast, consider what happens as $v_t \rightarrow \bar{v}$. By (15), in the limit,

$$x_t \approx (\bar{\delta}\beta\bar{v})^{-\frac{1}{\beta+\gamma-1}} A_t^{-\frac{\alpha+\gamma-1}{\beta+\gamma-1}}. \quad (27)$$

At large t , permanently multiplying A by $m > 1$ multiplies x_s , at each $s \geq t$, by approximately $m^{-\frac{\alpha+\gamma-1}{\beta+\gamma-1}}$. In conjunction, the increase to A_s and the proportional decrease to x_s multiply δ_s by $m^{-\frac{(\beta-\alpha)(\gamma-1)}{\beta+\gamma-1}}$ for $s \geq t$. Similarly, permanently dividing ρ by $m > 1$ multiplies x_s ($s \geq t$) by approximately $m^{-\frac{1}{\beta+\gamma-1}}$, which multiplies δ_s ($s \geq t$) by approximately $m^{-\frac{\beta}{\beta+\gamma-1}}$. The impacts are equal iff

$$\begin{aligned} (\beta - \alpha)(\gamma - 1) &= \beta \\ \iff \gamma &= 2 + \frac{\alpha}{\beta - \alpha}, \end{aligned} \quad (28)$$

with the level effect more impactful if the left-hand side is greater and the decrease to ρ more impactful if the right-hand side is greater. The growth-based intervention is more impactful when γ is higher, because higher values of γ motivate faster transitions from consumption to risk-reduction.

Since $\beta > \alpha > 0$, expression (28) reveals that the level effect can only be more impactful in this model if $\gamma > 2$. Still, it is notable that mere level effects to growth can ultimately affect the probability of survival at a comparable scale to permanent, equally-proportioned decreases to the social rate of pure time preference (holding technology growth fixed). Put another way, even temporary stagnation can carry long-term costs similar to those of permanently moving ethical attitudes away from concern for the future.

3.5 Generalization

The results of Sections 3.3 and 3.4 are set in the economic environment of Section 3.2. The three central ingredients of this environment are of course the growth path of technology, the hazard rate as a function of technology and policy, and the utility function. A particular functional form is assumed for each.

Throughout this section we will maintain the assumption of a CRRA utility function with $\gamma > 1$. We will however greatly relax our assumptions on the technology path and the hazard rate, to identify precisely the conditions under which the central lessons of Sections 3.3–3.4 are maintained.

In Sections 3.5.2–3.5.3, generalizing Proposition 1 from Section 3.3, we find that growth motivates increasing concern for safety: it is often optimal to set $x = 1$ early in time and $x \rightarrow 0$ late in time. A central result is that, except in cases where lowering risk is so difficult that it is not achieved even with *stagnation in consumption*, the hazard rate is also driven to 0.

In Section 3.5.4, generalizing Section 3.4, we likewise find that when a hazard function is compatible with survival, faster growth in consumption technology generally increases the probability of survival. The results support the robustness of the primary lessons drawn from hazard function (5): that survival is likely possible on the optimal path, and that faster consumption technology growth, if optimally regulated, will raise its probability.

3.5.1 Assumptions

Assumptions on technology growth

Instead of assuming that technology grows exponentially, we will assume only that $A_{(\cdot)}$ satisfies the following conditions:

- A1. right-continuous differentiability with $\dot{A}_t > 0$ for all t ;
- A2. $A_0 > 1$;
- A3. $\lim_{t \rightarrow -\infty} A_t = 0$; and
- A4. $\lim_{t \rightarrow \infty} A_t = \infty$.

We will call a technology path $A_{(\cdot)}$ admissible if it satisfies A1–A4.

We will continue to treat the growth of technology as exogenous. Importantly, this is without loss of generality for our purposes. We are concerned with the implications of a given technology path for optimal policy and cumulative risk, not with how a given technology path is generated. We therefore do not need to model how the rate of technological development may itself be determined by a planner’s funding of research and development, innovation by market participants, or any other forces.

Assumptions on the hazard rate

We will also consider a wider class of hazard functions. Among these, we will find relatively simple conditions under which a given hazard function and a given technology growth path are compatible with survival on the planner-optimal policy.

Return to the three desiderata preceding the introduction of hazard function (5). We will assume weakenings of two of these desiderata directly, and certain results will require a weakening of the third. In particular, we will universally assume that the hazard rate increases in x no less quickly than in A and is weakly convex in x . For certain results we will assume that the hazard rate does not decrease too quickly in A .

We will add to these the preliminary conditions that $\delta(\cdot)$ is continuously differentiable; that, when consumption equals zero, so that the entire productive capacity of society is dedicated to existential risk reduction, $\delta = 0$; and that otherwise $\delta > 0$.¹⁷

Formally, we will assume at most that the hazard rate is a function of $A > 0$ and $x \in (0, 1]$ satisfying the following conditions:

- D1. $\delta(A, x) > 0$,
- D2. $\lim_{x \rightarrow 0} \delta(A, x) = \lim_{A \rightarrow 0} \delta(A, x) = 0$,
- D3. twice continuous differentiability,¹⁸
- D4. $\eta_x(A, x) \geq \eta_A(A, x)$, and
- D5. weak concavity in x ,

where η_y denotes the elasticity of δ with respect to $y \in \{A, x\}$. We will call a hazard function admissible if it satisfies D1–D5.

Note that the constant elasticity hazard function of Sections 3.2–3.4 is admissible, with $\eta_A = \alpha$ and $\eta_x = \beta$ independent of A and x . Note also that we do not require $\eta_A(A, x)$ always to be positive. That is, we will allow for the possibility that new technologies sometimes lower the hazard rate at a given degree of foregone consumption.

¹⁷Recall that the hazard rate denotes the flow probability of *anthropogenic* existential catastrophe.

¹⁸We will define $\frac{\partial \delta}{\partial y}(A, 1) \equiv \lim_{x \rightarrow 1} \frac{\partial \delta}{\partial y}(A, x)$ for $y \in \{A, x\}$, and allow these derivatives to be infinite.

3.5.2 The end of consumption growth

Let $C^* \equiv \lim_{t \rightarrow \infty} A_t x_t$, when this limit is defined.

Given hazard function (5), $C^* = \infty$. This follows from Proposition 1. Since $\lim_{t \rightarrow \infty} g_{xt} = -\frac{\alpha+\gamma-1}{\beta+\gamma-1}g$, $\lim_{t \rightarrow \infty} g_{Ax,t} = (1 - \frac{\alpha+\gamma-1}{\beta+\gamma-1})g$, which is positive by the assumption that $\beta > \alpha$. In the long run, consumption growth is positive and exponential.

However, some admissible hazard functions motivate decreases to x fast enough that we do not have $C^* = \infty$. C^* may be finite, or C_t may oscillate indefinitely without growing ever higher.

Proposition 3. *The end of consumption growth*

Given an admissible hazard function $\delta(\cdot)$, define

$$\begin{aligned} R(C) &\equiv \lim_{A \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A, \frac{C}{A} \right) \frac{C^\gamma}{A} \bar{v}, \\ R^* &\equiv \lim_{C \rightarrow \infty} R(C). \end{aligned} \tag{29}$$

Given an admissible technology path and hazard function,

- a) *If $R^* \leq 1$, then $C^* = \infty$.*
- b) *If $R^* > 1$, then $C^* \neq \infty$.*

Proof. See Appendix B.3. □

To interpret the result, recall that $x = C/A$. The limit in (29) characterizes, if C is fixed even as A grows, what happens to the ratio of the marginal value of lowering x via increased safety ($\frac{\partial \delta}{\partial x} \cdot v$) to the marginal utility of raising x via increased consumption ($AC^{-\gamma}$). If the ratio approaches 1, then it is optimal for consumption to stagnate in the long run at C . If the ratio is greater than 1 for sufficiently large C , therefore, then stagnation at some finite C is optimal. If the ratio is less than or equal to 1 even as $C \rightarrow \infty$, then stagnation is not optimal.

Recall from (13) that $\bar{v} \equiv \frac{1}{\rho(\gamma-1)}$. When $R(C) > 0$, therefore, $R(C)$ is decreasing in ρ . A lower discount rate ρ can thus shift R^* from below to above 1, resulting in stagnation when there would otherwise have been long-run consumption growth, but never the reverse. There is no general result that consumption stagnation is desirable when ρ is sufficiently low, or undesirable when ρ is sufficiently large: for many hazard functions, as implicitly shown

at the end of Section 3.5.3, R^* is above 1 (even infinite) or below 1 (even 0) for any $\rho > 0$. Still, Proposition 3 illustrates how calls for an “end to growth” of some kind may be compatible with the results at the heart of this paper. Concern for the future can motivate controls on technological deployment strict enough to halt growth in *consumption*, despite the tendency for accelerating (even impatiently-regulated) *technological development* to lower cumulative risk.

3.5.3 The Kuznets curve generalized

Proposition 4. *The Kuznets curve generalized*

Given an admissible technology path and hazard function,

- a) $\lim_{t \rightarrow -\infty} x_t = 1$.
If η_A is bounded above $1 - \gamma$, then $\lim_{t \rightarrow \infty} x_t = 0$.
- b) $\lim_{t \rightarrow -\infty} \delta_t = 0$.
If $C^* = \infty$, then $\lim_{t \rightarrow \infty} \delta_t = 0$.
If $C^* \neq \infty$, η_A is bounded above $1 - \gamma$, and η_x is upper-bounded, then $\lim_{t \rightarrow \infty} \delta_t \neq 0$.

Proof. The proof of (a) is given in Appendix B.4. The proof of (b) is as follows.

By D1, D2, and D5, $\delta(A, x)$ is non-decreasing in x . So for all t , $\delta_t \leq \delta(A_t, 1)$. By D2, $\lim_{A \rightarrow 0} \delta(A, 1) = 0$. So by A3, $\lim_{t \rightarrow -\infty} \delta_t = 0$.

For the positive limit, begin with the weak first-order condition that the marginal flow utility of increasing x must weakly exceed the marginal cost via an increased hazard rate. Then multiply both sides by x_t :

$$\begin{aligned} A_t^{1-\gamma} x_t^{-\gamma} &\geq \frac{\partial \delta}{\partial x}(A_t, x_t) v_t \\ \implies (A_t x_t)^{1-\gamma} &\geq \frac{\partial \delta}{\partial x}(A_t, x_t) x_t v_t. \end{aligned} \tag{30}$$

If $C^* = \infty$, the left-hand side of (30) tends to 0. Since v is (eventually) positive and does not fall by D4, $\frac{\partial \delta}{\partial x} x \rightarrow 0$. Since $\frac{\partial \delta}{\partial x} x \geq \delta$ by D1 and D5, $\delta \rightarrow 0$.

If η_A is bounded above $1 - \gamma$, $\lim_{t \rightarrow \infty} x_t = 0$ by (a). Since eventually $x_t < 1$, eventually (30) holds with equality. If $C^* \neq \infty$, the left-hand side does not tend to 0 in the limit. Because v_t is upper-bounded, $\frac{\partial \delta}{\partial x} x$ does not tend to zero either. So if $\eta_x \equiv \frac{\partial \delta}{\partial x} \frac{x}{\delta}$ is upper-bounded, $\delta \not\rightarrow 0$. $\square$

Part (b) of the proposition stems from the fact that, as long as consumption rises without bound, its marginal utility falls to zero. If the hazard rate does not also fall to zero, the marginal value of sacrificing consumption to lower it further stays positive. The hazard rate must therefore fall to zero.

Even so, unbounded consumption growth does not necessarily coincide with a positive probability of survival. To achieve $S_\infty > 0$, δ_t must not only fall to 0 but fall sufficiently quickly. This in turn is guaranteed whenever consumption rises sufficiently quickly, which holds under a strengthening of the condition for unbounded consumption growth from Proposition 3.

Proposition 5. *Survival generalized*

Given an admissible hazard function $\delta(\cdot)$ and an admissible technology path $A_{(\cdot)}$ such that, for some $k > 1$ and some $\underline{t}$ we have

$$A_t \geq t^{\frac{k}{\gamma-1}} \quad \forall t > \underline{t}, \quad (31)$$

define

$$\tilde{R}(k) \equiv \lim_{t \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_t, \frac{t^{\frac{k}{\gamma-1}}}{A_t} \right) \frac{t^{\frac{k\gamma}{\gamma-1}}}{A_t} \bar{v}.$$

- a) *If $\lim_{k \downarrow 1} \tilde{R}(k) < 1$, then $\exists \underline{t} : C_t > t^{\frac{1}{\gamma-1}} \quad \forall t > \underline{t}$ and $S_\infty > 0$.*
- b) *If $\lim_{k \uparrow 1} \tilde{R}(k) > 1$, then $\exists \underline{t} : C_t < t^{\frac{1}{\gamma-1}} \quad \forall t > \underline{t}$.
If in addition η_x is upper-bounded, then $S_\infty = 0$.*

Proof. See Appendix B.5. □

Observe that, similar to $R(\cdot)$, $\tilde{R}(k)$ is the long-run ratio of the marginal value of lowering risk to the marginal value of increasing consumption when

$$C_t \propto t^{\frac{k}{\gamma-1}}. \quad (32)$$

If $\tilde{R}(k) < 1$ on this consumption path, for some $k > 1$, then on this path consumption grows too slowly. It is eventually preferable to raise x_t above its implied level of approximately $t^{\frac{k}{\gamma-1}}/A_t$. So if $\lim_{k \downarrow 1} \tilde{R}(k) < 1$, C_t eventually grows more quickly than (32) for some $k > 1$ on the optimal path. Conversely, if $\lim_{k \uparrow 1} \tilde{R}(k) \geq 1$, C_t eventually grows more slowly than (32) for $k = 1$.

If C_t grows more quickly than (32) for some $k > 1$, then the left-hand side of (30) falls more quickly than t^{-k} for some $k > 1$. So $\frac{\partial \delta}{\partial x}x$ does as well. Recalling that $\delta < \frac{\partial \delta}{\partial x}x$, this ensures a positive probability of survival.

If C_t grows more slowly than (32) for $k = 1$, then the left-hand side of (30) falls more slowly than $1/t$. The right-hand side equals $\frac{\partial \delta}{\partial x}x \cdot v = \eta_x/\delta \cdot v$. If η_x is upper-bounded, δ falls more slowly than $1/t$. Cumulative risk is therefore infinite, and survival is impossible.

For illustration, let us evaluate the constant elasticity hazard function of Section 3.2 for the case of exponential growth at rate g .

$$\begin{aligned}\tilde{R}(k) &= \lim_{t \rightarrow \infty} \bar{\delta} e^{\alpha g t} \beta \left(\frac{t^{\frac{k}{\gamma-1}}}{e^{g t}} \right)^{\beta-1} \frac{t^{\frac{k\gamma}{\gamma-1}}}{e^{g t}} \bar{v} \\ &= \bar{\delta} \beta \bar{v} \lim_{t \rightarrow \infty} e^{-(\beta-\alpha)g t} t^{\frac{\beta+\gamma-1}{\gamma-1}k} = 0\end{aligned}\tag{33}$$

for any k , since $\beta > \alpha$. So $\lim_{k \downarrow 1} \tilde{R}(k) = 0 < 1$. Part (a) of Proposition 5 thus generalizes the conclusion of footnote 16 that, with hazard function (5), consumption grows at least as quickly as a sufficient power function (in fact it grows exponentially) and that there is a positive probability of survival.

By contrast, consider the constant elasticity hazard function but with $\alpha = \beta$. In this case, (33) = ∞ for any k , so $\lim_{k \uparrow 1} \tilde{R}(k) = \infty > 1$. Also, η_x is constant at β , and so upper-bounded. $\delta(A, x) = Ax$ is thus an example of a hazard function satisfying D1–D5 for which the probability of survival on the optimal path is zero given exponential technology growth (and indeed given any $A_{(\cdot)}$ that is eventually bounded above zero).

3.5.4 Acceleration and risk generalized

For any admissible hazard function, the lessons of Section 3.4 are essentially maintained. The effect of a temporary level effect on the probability of survival is ambiguous, as detailed in Appendix A.2. However, if the probability of survival is positive on the planner-optimal policy path, given the baseline technology path, then an acceleration to technological development increases the probability of survival. If the probability of survival is zero on the planner-optimal policy path, then an acceleration to technological development may increase the probability of survival or have no effect. We will now state these results more generally and precisely.

Choose an admissible technology path $A_{(\cdot)}$ and hazard function $\delta(\cdot)$.

A differentiable path $\tilde{A}_{(\cdot)}$ is an acceleration from $\underline{A} \in [A_0, \infty)$ to $\bar{A} \in (A_t, \infty]$ if $\tilde{A}_0 = A_0$ and

$$\begin{aligned}\dot{\tilde{A}}_A &= \dot{A}_A, & A < \underline{A}; \\ &\geq \dot{A}_A, & A = \underline{A}; \\ &> \dot{A}_A, & A \in (\underline{A}, \bar{A}); \\ &= \dot{A}_A, & A \geq \bar{A}.\end{aligned}$$

The acceleration is permanent if $\bar{A} = \infty$ and temporary otherwise.

Let $\tilde{A}_{(\cdot)}$ be an acceleration from $\underline{A}$. At $A < \underline{A}$, $\tilde{v}_A$ is the costate variable on survival, i.e. the expected value of the future, at technology level A given that the subsequent technology path follows $A_{(\cdot)}$. At $A \geq \underline{A}$, $\tilde{v}_A$ is the costate variable on survival at technology level A given that the subsequent technology path follows $\tilde{A}_{(\cdot)}$. Then

- $\tilde{x}_A$ is given by (15) with $A, \tilde{v}_A$ in place of A_t, v_t .
- $\tilde{\delta}_A \equiv \delta(A, \tilde{x}_A)$.
- $\tilde{X} \equiv \int_{A_0}^{\infty} \tilde{\delta}_A \dot{\tilde{A}}_A^{-1} dA$.

Given a baseline technology level $\underline{A}$ and a technology growth rate $\dot{\tilde{A}} > \dot{A}_{\underline{A}}$, denote by $\tilde{A}_{(\cdot)}[\epsilon]$ the acceleration from $\underline{A}$ to $\underline{A} + \epsilon$ with¹⁹

$$\dot{\tilde{A}}_A = \dot{\tilde{A}}, \quad A \in [\underline{A}, \underline{A} + \epsilon).$$

Then the effect on cumulative risk, per unit of technological development, of instantaneously accelerating to $\dot{\tilde{A}}$ at $\underline{A}$ is defined to be

$$\Delta_{\underline{A}, \dot{\tilde{A}}} \equiv \lim_{\epsilon \rightarrow 0} (\tilde{X}[\epsilon] - X)/\epsilon,$$

where $\tilde{X}[\epsilon]$ is cumulative risk $\tilde{X}$, as defined above, given acceleration $\tilde{A}_{(\cdot)}[\epsilon]$. (The effect on an instantaneous acceleration on cumulative risk *per unit time* is $\Delta_{\underline{A}, \dot{\tilde{A}}} \dot{\tilde{A}}$, since during the acceleration, $\dot{\tilde{A}}$ units of technology are developed per unit time. This is of course of the same sign.)

¹⁹For some $\bar{\epsilon} > 0$ this is indeed an acceleration for all $\epsilon < \bar{\epsilon}$, by assumption A1 that $A_{(\cdot)}$ is right-continuously differentiable.

Proposition 6. Acceleration weakly lowers risk

Choose an admissible technology path $A_{(\cdot)}$ and hazard function $\delta(\cdot)$.

Given $\underline{A}$, $\dot{\tilde{A}}$ with $\dot{\tilde{A}} > \dot{A}_{\underline{A}}$,

$$\text{a) } \Delta_{\underline{A}, \dot{\tilde{A}}} = \delta_{\underline{A}}(\dot{\tilde{A}}^{-1} - \dot{A}_{\underline{A}}^{-1}) < 0.$$

Given an acceleration $\tilde{A}_{(\cdot)}$ from $\underline{A}$ to $\bar{A}$,

$$\text{b) } \text{If } X < \infty, \text{ then } \tilde{X} \leq X + \int_{\underline{A}}^{\bar{A}} \Delta_{\underline{A}, \dot{\tilde{A}}} dA < X.$$

- c) If $X = \infty$ and the acceleration is temporary, then $\tilde{X} = \infty$.
 If $X = \infty$ and the acceleration is permanent, then $\tilde{X}$ may be finite or infinite.

Proof. See Appendix B.6. □

The intuition is the same as illustrated in Section 3.4. Acceleration in effect horizontally rescales all or part of the hazard curve by leaving less time spent at each state. It may also induce more stringent policy at each state, in which case the weak inequality of part (b) is strict.

Accelerations vs. level effects

Given a technology path $A_{(\cdot)}$ satisfying A1 and A4, say that a differentiable technology path $\tilde{A}_{(\cdot)}$ is a level effect to $A_{(\cdot)}$ (at time 0) if

$$\exists m > 1 : \tilde{A}_t = mA_t \quad \forall t.$$

When technology growth is exponential, level effects are (sharp) temporary accelerations. Otherwise, they may be distinct.

Unlike temporary accelerations, level effects do not always decrease cumulative risk outside the exponential growth context. Consider for example hazard function (5) with a technology path $A_{(\cdot)}$ that is nearly stagnant for an arbitrarily long period, say for $t \leq 99$; that grows exponentially at $t > 99$; and for which the implied regime-change time is $t^* = 100$. A level effect—a jump in the technology level at $t = 0$ —then raises the technology level during the arbitrarily long period of stagnation, which non-negligibly raises cumulative risk, while lowering cumulative risk only negligibly by cutting a vertical slice from the hazard curve following $t = 99$.

The direction of technical change

At face value, this is a model in which there is a single dimension to technological development. Inventions simply occur in sequence, each of which increases potential consumption but has an idiosyncratic effect on the hazard rate at any given level of consumption. (Recall that we allow $\delta(A, x)$ to decrease in A .) This one-dimensionality may seem unrealistic. In practice, technological development is surely at least somewhat *directed*, with the tradeoffs between consumption and risk in later periods affected by the extent to which policymakers and market participants in earlier periods have supported research into various types of technology. Consider for example the “richer model” of Jones (2016), in which increases in the value of life relative to consumption motivate increases not only in health spending but also in medical R&D.

As with our assumption that the baseline growth rate of technology at each time is exogenous, however, the assumption that the path of technology is exogenous is also essentially without loss of generality. A path of maximum potential consumption levels $\{A_t\}$ and a hazard function $\delta(A, \cdot)$ simply describe a path of possibilities frontiers over time, without embedding any assumptions about how this path of possibilities frontiers is generated. If we posit a wider space of possible production technologies than the sequence adopted on the baseline path, we must clarify that “accelerations” consist of increases to the rate of motion along the baseline path.

Proposition 6 only applies to accelerations in this sense. Subsidizing the development of risky technologies that would not otherwise have been invented, or choosing a technology path on which they are invented sooner than they would have been but risk-decreasing technologies are not, does not necessarily lower cumulative risk.²⁰

²⁰In addition to modeling the policy choice about how much consumption to sacrifice for an instantaneous reduction to the hazard rate, an earlier version of this paper models the technology path as directed by policy as well. The growth model is semi-endogenous, so total potential technology growth is driven by exogenous population growth, but research is optimally allocated between risk-increasing “consumption technology” and risk-decreasing “safety technology”. Conceptually, that model sheds light on the same question as this one—how acceleration affects cumulative risk, given an endogenous policy response—but the objects of study are accelerations to population rather than to technology itself.

Numerical estimation suggests that acceleration weakly decreases cumulative risk in that context as well, for the same reasons as it does here. When population growth is accelerated, and labor is allocated optimally across fields, civilization traverses roughly the

In Appendix A.3, the lessons of this generalized model are used to explore two particular hazard functions that may be of interest. The first illustrates that, early in time, the hazard rate may increase alongside smooth declines in x . The second is “microfounded” by an assumption that increases in safety expenditure lower risk through redundant safeguards.

4 Transition risk

4.1 Motivation

A hazard function of the form $\delta(A_t, x_t)$ captures what we have called “state risk”: δ depends on the *level* of technology. On this framing, it is perhaps unsurprising that escaping risky states more quickly lowers cumulative risk.

But risk may instead be “transitional”: posed by *technological development*. This is the intuition captured by Jones’s (2016) “Russian roulette” model of technological development and (2024) model of AI risk, and by Bostrom’s (2019) analogy to drawing potentially destructive balls from an urn. Perhaps stagnation at a given level of technology is essentially safe, and risk arises in the process of discovering and deploying new technologies with unknown consequences. If so, given a positive-growth baseline, does accelerating technological development further increase cumulative risk?

4.2 A transition-risk-based hazard function

To explore this possibility, suppose δ increases in $\dot{A}_t$ instead of, or as well as, in A_t . For simplicity, we will again restrict our consideration to a constant elasticity hazard function:

$$\delta_t = \bar{\delta} A_t^\alpha \dot{A}_t^\zeta x_t^\beta, \quad \bar{\delta} > 0, \zeta \geq 0, \beta > 1. \quad (34)$$

Our original hazard function (5) is the special case of (34) with $\zeta = 0$ (and $\beta > \alpha > 0$). This model is thus an alternative generalization of hazard function (5), complementary to that of Section 3.5.

same technology path but more quickly. Furthermore, when future population growth is anticipated to be faster, the value of the future is higher (due to faster future technological development even if larger populations are not valued more intrinsically), so optimal policy shifts the technology path in a safer direction.

As long as $\zeta > 0$, however, the model is most straightforwardly interpreted as one in which new technologies—new “draws from Bostrom’s urn”—consist of absolute increases to A . Holding policy fixed, the introduction of multiple technologies can pose more, less, or equal risk if they are introduced concurrently than if they are introduced in sequence, depending on whether ζ is greater than, less than, or equal to 1. The development of more advanced technologies can pose more, less, or equal risk as compared to the development of less advanced technologies, depending on the sign of α .

Alternatively, the model may be interpreted as one in which new technologies consist of proportional increases to A . This can be seen by rewriting the hazard function as

$$\delta_t = \bar{\delta} A_t^{\alpha+\zeta} \left(\frac{\dot{A}_t}{A_t} \right)^\zeta x_t^\beta.$$

On this interpretation, the assumption that $\alpha + \zeta > 0$ amounts to the assumption that the development of more advanced technologies poses weakly more risk than the development of less advanced technologies. Because $\dot{A}/A$ has been approximately constant throughout the last century, the view that the hazard rate has risen must be attributed to the increasing danger of each “technological development” in this sense.

Finally, consider the case of $\alpha = -1$, $\zeta = 1$, so that

$$\delta_t = \bar{\delta} \frac{\dot{A}_t}{A_t} x_t^\beta.$$

Here, fixing x , each proportional increase to A induces a constant hazard, independently of how quickly the increase occurs. In the absence of policy—with $x = 1$ (or any other constant) permanently—this model is essentially equivalent to the “Russian roulette” model of Jones (2016)²¹ and the AI risk model of Jones (2024).

4.3 Acceleration and transition risk

Without mitigation

Suppose that the baseline technology path $A_{(\cdot)}$ satisfies A1. We will not impose A4, that the path grows without bound; we will let $\hat{A} \equiv \lim_{t \rightarrow \infty} A_t$ be finite or infinite.

²¹Our $\bar{\delta}$ is the variable there denoted π .

As implied above, fixing policy, whether acceleration increases or decreases cumulative risk depends on whether ζ is greater or less than 1. This can, as usual, be seen most clearly by integrating the hazard curve with respect to A :

$$X = \int_0^\infty \bar{\delta} A_t^\alpha \dot{A}_t^\zeta dt = \int_{A_0}^{\hat{A}} \bar{\delta} A^\alpha \dot{A}_A^{\zeta-1} dA.$$

Given acceleration $\tilde{A}_{(\cdot)}$ from $\underline{A} \in [A_0, \hat{A})$ to $\bar{A} \in (\underline{A}, \hat{A}]$, cumulative risk equals

$$\tilde{X} = X + \int_{\underline{A}}^{\bar{A}} \bar{\delta} A^\alpha (\dot{\tilde{A}}^{\zeta-1} - \dot{A}_A^{\zeta-1}) dA.$$

The integral is negative if $\zeta < 1$, zero if $\zeta = 1$, and positive if $\zeta > 1$.

In the Russian roulette model, for instance, though there is a technology level $\hat{A} < \infty$ at which it is optimal to halt technological development (see Appendix A.4), accelerating technological development before $\hat{A}$ does not affect cumulative risk.

With mitigation

In Section 2, we saw that acceleration weakly lowered cumulative state risk in the absence of policy, and in Section 3, we saw that the tendency of acceleration to lower cumulative state risk was amplified by the presence of optimal policy. Here, we have seen that the impact of acceleration on cumulative transition risk is ambiguous in the absence of policy. We will now see that it remains ambiguous in the presence of optimal policy, but that policy can again introduce a tendency for acceleration to lower cumulative risk.

For simplicity, we will now again impose the assumption that A grows at a constant exponential rate g . Also, since given exponential growth $g_{\hat{A}} = g$, we will impose

$$\beta > \alpha + \zeta, \tag{35}$$

which, rather than $\beta > \alpha$, is now the condition necessary for survival without $C_t = A_t x_t \rightarrow 0$. Under these conditions, since $\dot{A}$ is proportional to A , the planner's problem is precisely as described in Section 3.3, with $\alpha + \zeta$ taking

the place of α (up to a coefficient g^ζ that can be rolled into $\bar{\delta}$). Baseline x and δ paths, and S_∞ , are unchanged. The existential risk Kuznets curve remains.

Let A^* denote the uppermost technology level at which it is optimal to set $x = 1$ on the baseline technology path. Since the first-order condition

$$\begin{aligned} \frac{\partial u}{\partial x_t}(A_t, x_t) &\geq \frac{\partial \delta}{\partial x_t}(A_t, x_t)v_t \\ \implies A_t^{1-\gamma}x_t^{-\gamma} &\geq \bar{\delta}A_t^\alpha\beta x_t^{\beta-1}v_t \end{aligned}$$

must be satisfied everywhere and hold with equality for $x < 1$, we have

$$x_A = \begin{cases} 1 & A \leq A^*, \\ \left(\bar{\delta}\beta A^{\alpha+\gamma-1}\dot{A}_A^\zeta v_A\right)^{-\frac{1}{\beta+\gamma-1}} & A > A^*. \end{cases} \quad (36)$$

Substituting (36) into the expression for cumulative risk

$$X = \int_{A_0}^{\infty} \bar{\delta} A^\alpha \dot{A}_A^{\zeta-1} x_A^\beta dA, \quad (37)$$

we have

$$\begin{aligned} X &= \int_{A_0}^{A^*} \bar{\delta} A^\alpha \dot{A}_A^{\zeta-1} dA \\ &+ \int_{A^*}^{\infty} \left(\bar{\delta}^{1-\gamma}\beta^\beta A^{(\beta-\alpha)(\gamma-1)}v_A^\beta\right)^{-\frac{1}{\beta+\gamma-1}} \dot{A}_A^{\zeta\frac{\gamma-1}{\beta+\gamma-1}-1} dA. \end{aligned} \quad (38)$$

Recall that a technology path $\tilde{A}_{(\cdot)}$ is an acceleration if $\dot{\tilde{A}}_A > \dot{A}$ for technology levels $\underline{A} \in [A_0, \infty)$ to $\bar{A} \in (\underline{A} \leq \infty]$. With or without policy, an acceleration affects cumulative risk directly, by changing the technology growth rate from $\underline{A}$ to $\bar{A}$. In the presence of a policy response, an acceleration also affects cumulative risk indirectly by affecting v_A for $A \in [\underline{A}, \bar{A})$, which affects the policy response undertaken at the given range of technology levels.

Under the hazard functions of the previous sections, as we have seen, faster technology growth is always weakly preferred. This follows from the fact that it is feasible to offset higher values of A_t with lower choices of x_t , such that the original consumption path is maintained, and from assumption

D4 that given this policy response, the hazard curve is weakly lowered. Since a future with faster growth is more valuable, an acceleration from $\underline{A}$ to $\overline{A}$ raises v_A for $A \in [\underline{A}, \overline{A})$.

Under hazard function (34), this argument is no longer valid. This is because, unlike an increase to A_t , an increase to $\dot{A}_t$ brings no contemporaneous benefit; yet the contribution of faster technology growth to risk can still be mitigated only with a lower policy choice and thus less contemporaneous consumption. And indeed, under hazard function (34), faster technology growth is no longer always preferred. We can see this most straightforwardly in the case of $\alpha = -1$, $\zeta = 1$: as noted above, this is the Russian roulette model of Jones (2016), and as Jones finds, with $\gamma > 1$, it is optimal for technology to grow only to a finite level. In the more general model here, the result that stagnation is optimal is knife-edge, as discussed in Appendix A.4. Nevertheless, the result that an acceleration from $\underline{A}$ to $\overline{A}$ does not necessarily yield $\tilde{v}_A \geq v_A$ for $A \in [\underline{A}, \overline{A})$ holds more generally.

These complexities are avoided when we focus on instantaneous accelerations. The impact of an acceleration from $\underline{A}$ to $\overline{A}$ on v_A , for $A \in [\underline{A}, \overline{A})$, falls to zero as $\overline{A} - \underline{A} \rightarrow 0$. The impact of a *brief* acceleration on cumulative risk is therefore approximately the impact found when we ignore impacts on v_A .

Proposition 7. *Instantaneous acceleration and transition risk*

Given hazard function (34) and technology path (4), choose a technology level $\underline{A} > 1$ and growth rate $\dot{A} > \dot{A}_{\underline{A}}$. If

- a. $\underline{A} \geq A^*$ and $\zeta < (=, >) 1 + \frac{\beta}{\gamma-1}$, or
- b. $\underline{A} < A^*$, $\zeta < (=, >) 1$, and $\dot{A}$ maintains (36) = 1 at $A = \underline{A}$,

then $\Delta_{\underline{A}, \dot{A}} < (=, >) 0$.

Proof. See Appendix B.7. □

The result follows essentially immediately from the exponent on $\dot{A}_A$ in (37). In particular, instantaneous acceleration after A^* lowers cumulative risk as long as

$$\zeta \frac{\gamma - 1}{\beta + \gamma - 1} - 1 < 0 \implies \zeta < 1 + \frac{\beta}{\gamma - 1}. \quad (39)$$

It is sufficient, though not necessary, for (39) that

$$\zeta \leq 1 \quad \text{or} \quad \alpha \geq -1, \gamma \leq 2.$$

The $\zeta \leq 1$ case follows from the fact that $\frac{\gamma-1}{\beta+\gamma-1} < 1$. The $\alpha \geq -1, \gamma \leq 2$ case follows from the fact that if $\alpha \geq -1$, then, by (35), $\zeta < \beta + 1$, so $\frac{\zeta}{\beta+1} < 1$. Since macroeconomic estimates of $\gamma \leq 2$ are standard, this result suggests that accelerations to technology growth lower cumulative risk on the optimal path in the context of transition risk, at least if they occur late enough in time that mitigation is already underway.

Furthermore, this is, again, even without considering the fact that an increase to future growth can change the value of the future. Though the direction of this change is in principle ambiguous, most observers today would take it for granted that, at least on a conventional discount rate, faster technology growth would be a benefit on the current margin. This would then be another channel through which a (positive-duration) acceleration would motivate greater concerns today for safety.

It may be counterintuitive that instantaneous acceleration reduces risk only when γ lies *below* a bound, because higher values of γ lower the marginal utility of consumption and motivate more rapid reallocations of resources from consumption to safety. The result is driven by the fact that, when γ is high, the marginal utility of consumption rises rapidly as x is cut, so only a small cut to x suffices to maintain the condition that the marginal utility of consumption equals the marginal disutility incurred by raising the hazard rate. The higher γ is, the more quickly x falls as A rises, but the *less* sensitive x is to a change in $\partial\delta/\partial x$ —e.g. an increase due to higher $\dot{A}$ —at a given value of A .

The nonrivalry of safety effort

Hazard function (34) is explored here mainly for its simplicity and similarity to (5).

One valid criticism of this functional form is that it overemphasizes a channel through which the risks posed by a series of technological developments can be cheaper to mitigate if they occur at once than if they occur in sequence. Suppose that $\beta \approx 1$, that $\zeta = 1$, and that two small increases to $\dot{A}$ —let us call them two “experiments”—can occur in sequence or simultaneously. If they occur in sequence, halving the risk posed by each requires

halving x and thus consumption for two periods in a row. If the experiments occur simultaneously, the same reduction in cumulative risk only requires halving consumption once.

For some kinds of experiments and some kinds of safety infrastructure, the assumption that safety efforts are “nonrival” in this sense is reasonable. Wastewater monitoring for the sake of early pandemic detection reduces the risk posed by potentially pandemic-inducing biological experiments by a proportion independent of how many experiments are underway.

In other cases, however, it is not reasonable. It does not apply, for instance, to the costs of the safety equipment that must be used at each lab. Safety efforts of this kind might be better modeled by a modified version of the “safety in redundancy” model of Appendix A.3.2.

A thorough attempt to shed light on the relationship between growth and transition risk would require further study. Nevertheless, the basic model explored here offers two lessons. First, in the absence of policy, the effect of acceleration on transition risk is ambiguous, and there is no effect in the arguably central “ $\zeta = 1$ ” case assumed by Jones (2016, 2024). Second, the presence of an optimal policy response can shift the relevant “ ζ ” threshold, in particular significantly shifting it upward to the extent that safety efforts are nonrival across contemporaneous risks.

Stagnation vs. deceleration

When $\zeta > 0$, complete stagnation ($\dot{A} = 0$) is always the safest path of all. Nevertheless, we have seen with and without policy that given a positive growth rate, faster growth can decrease risk.

The key to this puzzle is that, given stagnation at $\hat{A}$, levels of $A > \hat{A}$ are never attained. Cumulative risk is therefore not (37) but (37) with the ∞ replaced with $\hat{A}$. Absent stagnation, however slow the growth rate, all levels of A are attained. The growth rate only determines the risk endured at each one. The direct cost of faster progress during a given range of A -values (higher risk per unit time, to the extent that $\zeta > 0$) is partially, and may be more than fully, outweighed by the fact that faster progress motivates more mitigation at each point in time, in combination with the now familiar fact that when progress is faster we do not linger in a given range of A -values as long.

5 Conclusion

Human activity can create or mitigate existential risks. The framework presented here illustrates that, under a wide array of conditions, existential risk satisfies the conditions that should be expected to give rise to a Kuznets curve. This observation offers a potential economic explanation for the claim by some prominent thinkers that humanity is in a critical “time of perils”. We may be economically advanced enough to be able to destroy ourselves, but not yet enough that we are willing to make large sacrifices for the sake of safety. If we are indeed living through the time of perils, reductions to existential risk today have a massive expected impact on the course of the long-term future.

At the same time, this framework highlights a channel through which some efforts intended to reduce existential risk may backfire. In the absence of policy, when risk is posed by the existence of advanced technologies, broad-based decelerations to technological development typically either worsen or have no impact on the odds of long-term survival. In the presence of an optimal policy response, even by a policymaker with little concern for the long-term future, this impact is magnified. The impact can be significant, with proportional consumption decreases having comparable impacts to proportional increases in the planner’s rate of time preference. In the extreme, permanent stagnation in a world of risky technology can make a catastrophe inevitable that might otherwise have been avoided.

This lesson comes with three caveats. First, it is in no sense an argument against regulating the use of risky technologies. Indeed, a primary channel explored here through which technological development lowers risk is that it hastens the day when regulation is severe. Some recent reactions against calls to heavily regulate AI, e.g. that of Andreessen (2023), might be read as expressing the view that our “ x ” should never be set far below one. If that is so, it is not for the reasons presented in this paper.

Second, when risk is posed by the existence of advanced technologies, the conclusion that an unregulated or optimally regulated acceleration weakly lowers cumulative risk appears relatively robust. When risk is posed by the *development* of advanced technologies, however, this conclusion is much weaker. It holds in the “transition risk” models of Jones (2016, 2024), where acceleration has no impact on cumulative risk in the absence of policy. Under slight modifications to these models, however, it may fail. Policy may strengthen the tendency for acceleration to weakly decrease cumulative risk,

as the simple model of Section 4 illustrates. It seems likely, however, that a more thorough literature on the question of transition risk, growth, and regulation would identify plausible models under which policy dilutes or overturns this tendency.

Third, in the cases where we have found that policy magnifies a negative link between acceleration and cumulative risk, we have relied on the assumption that policy is optimal. If it is not, then the impact of acceleration on cumulative risk may be reduced or even overturned, as illustrated in Section 3.1. In fact, Shulman and Thornley (2024) argue that the policy response to hazardous technologies to date has been far from optimal, even with respect to a conventional discount rate. The appropriate broad lesson to draw about the impact of policy on the relationship between acceleration and risk is only that, to the extent that the regulatory regime equates, or will eventually move toward equating, the marginal utility of consumption to the marginal expected discounted utility of safety expenditure, consumption-increasing technological development today has the unseen but potentially large benefit of speeding future safety efforts. For slowing technological development to lower cumulative risk, the policy inefficiency in question must be severe and lasting enough to outweigh this benefit.

In this light, further research on the nature of policy distortions around the regulation of hazardous technologies would be valuable. Exploring the long-term implications of other models of anthropogenic existential risk, and of optimal policy in the face of it—beyond the simple state- and transition-risk-based relationships explored here—could be valuable as well, so as to better characterize the scope of the result that optimally regulated acceleration weakly lowers cumulative risk. If plausible models are found under which the result is overturned, this will naturally pose important questions which can only be answered empirically. For now, however, the results presented here suggest that even those exclusively concerned with reducing cumulative existential risk should often cheer technological advances despite their short-term hazards, and advocate risk-reduction measures today only when they are sufficiently targeted and the costs to technological development are sufficiently small.

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Appendices

A Supplemental materials

A.1 Calibrating the elasticity of the hazard rate to safety expenditures

Shulman and Thornley (2024) estimate that well-targeted expenditures of \$400B over the next decade would reduce the probability of existential catastrophe over the next decade by at least 0.1% in absolute terms, from a baseline of 1.85%.

The scale of the magnitude of the risk is taken from Ord’s (2020, p. 167) educated guesses and may be disputed. However, an estimate of β depends only on the *proportion* by which a given consumption sacrifice will reduce the hazard rate. We will rely on Shulman and Thornley’s assessment that expenditures of \$400B would multiply the probability of existential catastrophe over the next decade by at most

$$1 - \frac{0.1\%}{1.85\%} \approx 0.946, \quad (40)$$

while remaining agnostic about the the magnitude of the probability. For instance, we are trusting their assessments of the extent to which disease monitoring expenditures would be able to prevent existentially hazardous anthropogenic pandemics by helping authorities to contain them early, while remaining agnostic about the probability per year that such a pandemic will arise.

Global consumption per year is currently approximately \$72.5T (World Bank, 2022). If real consumption grows at 2% per year and the relevant interest rate is 5% per year, the present value of global consumption over the next ten years is approximately $\$72.5T \times (1 - e^{-10(0.05-0.02)})/(0.05 - 0.02) \approx \$626.4T$. A sacrifice of \$400B = \$0.4T in today’s dollars over the next decade is thus a sacrifice that multiplies consumption by a fraction of

$$1 - \frac{0.4}{626.4} \approx 0.99936. \quad (41)$$

Given $x^\beta < 0.946$ at $x \approx 0.99936$, it follows that

$$\beta > \frac{\log(0.946)}{\log(0.99936)} \approx 86.7.$$

This exercise of course tells us nothing about whether it is reasonable to assume a constant-elasticity hazard function in general. If the Shulman and Thornley estimate is correct within three orders of magnitude, however, it does prove that the hazard function is currently convex over at least some range of feasible consumption levels. This follows immediately from the facts that (41) > (40) and that the hazard rate cannot be cut by a proportion greater than one.

A.2 State risk with mitigation: Instantaneous level effects

Let

$$\eta_{xy}(A_t, x_t) \equiv \frac{\partial}{\partial y} \left(\frac{\partial \delta}{\partial x}(A_t, x_t) \right) \cdot \frac{\frac{\partial \delta}{\partial x}(A_t, x_t)}{y_t},$$

for $y \in \{x, A\}$, denote the elasticity of $\partial \delta / \partial x$ with respect to y .

If $A_t^{1-\gamma} > \frac{\partial \delta}{\partial x}(A_t, 1)v_t$, so that $x_t = 1$ and the $x_t \leq 1$ constraint binds, then multiplying A_t by m slightly above 1 multiplies δ_t by approximately $m^{\eta_A(A_t, 1)} \geq 1$.

If the $x_t \leq 1$ constraint does not bind, so that (30) is maintained with equality as A_t rises, then multiplying A_t by m slightly above 1 has a direct impact and possibly an indirect impact on the hazard rate. The direct impact is again to multiply δ_t by approximately $m^{\eta_A(A_t, x_t)}$. The possible indirect impact is to affect the choice of x_t . Letting $\xi(A_t, x_t)$ denote the elasticity of chosen x to A around (A_t, x_t) , to maintain equality (30) as A_t varies we must have

$$\begin{aligned} \xi(A_t, x_t) &= \frac{1-\gamma}{\gamma} - \frac{1}{\gamma} \left(\eta_{xA}(A_t, x_t) + \xi(A_t, x_t)\eta_{xx}(A_t, x_t) \right) \\ \implies \xi(A_t, x_t) &= -\frac{\eta_{xA}(A_t, x_t) + \gamma - 1}{\eta_{xx}(A_t, x_t) + \gamma}. \end{aligned}$$

(Observe that v_t is unaffected by an instantaneous change to A_t and x_t .) If $\xi(A_t, x_t) > 0$ and $x_t = 1$, the marginal increase to A_t does not affect the chosen x_t . Otherwise, the overall elasticity of the hazard rate to A_t , in the context of an instantaneous level effect, is not $\eta_A(A_t, x_t)$ but

$$\eta_A(A_t, x_t) + \xi(A_t, x_t)\eta_x(A_t, x_t). \quad (42)$$

This is negative in the context of hazard function (5), yielding the earlier result that when $x < 1$, temporary level effects lower the hazard rate. Under the weaker conditions here, the sign of (42) is ambiguous. This is illustrated in Appendix A.3.1, with a hazard function under which, early in time, increases to A —even combined with increases to v —motivate such slow decreases to x that on balance the hazard rate rises.

A.3 State risk with mitigation: Two hazard functions of interest

We will assume throughout this section that technology growth is exponential at rate $g > 0$. We will study the optimal paths of the hazard rate given by two particular admissible hazard functions.

A.3.1 A lower Inada condition on safety

As shown in Section 3.3, given a constant elasticity hazard function, δ rises as long as it remains optimal to maximize consumption, and falls immediately once it becomes optimal to begin choosing sub-maximal consumption out of concern for safety. And as noted at the end of Section 3.3, this result is arguably at odds with the experience of the last century. We will therefore here explore how to tweak the hazard function so that the Kuznets curve is smoothed, and the policy choice variable falls even early in time while the hazard rate is still rising.

A constant elasticity hazard function generates a distinct pair of regimes for the same reason here as in Stokey (1998): because, when $x = 1$, marginal “safety expenditures”—decreases to x —produce only finite marginal benefits. That is, there is no “lower Inada condition on safety”. It is therefore optimal to maximize consumption until the marginal utility of consumption has fallen and the marginal value of existential risk reduction efforts have risen, as we have seen, and then at once to begin lowering x roughly exponentially. We will say that a hazard function exhibits a lower Inada condition on safety if $\lim_{x \rightarrow 1} \frac{\partial \delta}{\partial x} = \infty$. Under this condition, it is optimal to set $x_t < 1$ as long as $v_t > 0$: as long as civilization is worth preserving at all, some expenditures on existential risk reduction are worthwhile.

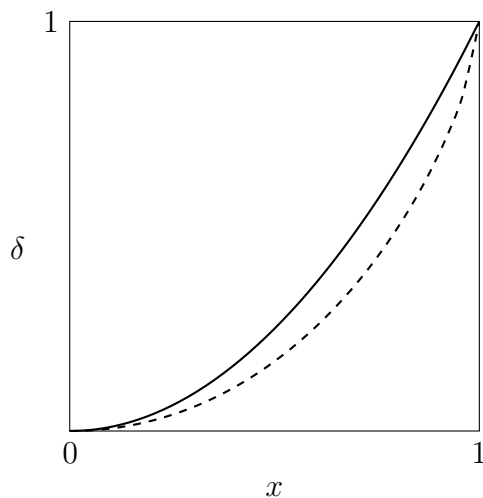
Not every hazard function with a lower Inada condition on safety behaves like a smoothed version of a constant elasticity hazard function. If the inverse of the hazard function is too concave around $x = 1$ (when A is low), then x

may fall rapidly, rather than mildly, from the outset, yielding no early period during which $x \approx 1$. If it is not concave enough around $x = 1$, on the other hand, then early decreases to x produce significant decreases to δ , so that the hazard rate falls even early in time.

One class of hazard functions with the desired features is

$$\delta_t = \bar{\delta} A_t^\alpha x_t^\beta \frac{1 - (1 - x_t)^\epsilon}{x_t}, \quad \epsilon \in \left(\frac{1}{2}, 1\right), \quad (43)$$

where the conditions on parameters other than ϵ are as before. The distinction between the hazard functions is illustrated below for the case of $\bar{\delta} A^\alpha = 1$, $\epsilon = 0.6$, $\beta = 2$. The solid curve represents the old hazard function; the dashed curve represents the new hazard function, vertical at $x = 1$.



Note that

$$\lim_{x \rightarrow 0} \frac{1 - (1 - x_t)^\epsilon}{x_t} = \epsilon,$$

so the asymptotics in this case are identical to those in the case of a constant elasticity hazard function (except that the hazard rate is multiplied by ϵ). The transition dynamics, however, are qualitatively different. Though it is now optimal to set $x < 1$ as long as $v > 0$, x now falls smoothly and δ smoothly rises and falls. The paths of the hazard rate and policy choice are

illustrated below for $\epsilon = 0.6$, $A_0 = 2.03$, and otherwise the same parameter values as in Table 1.²²

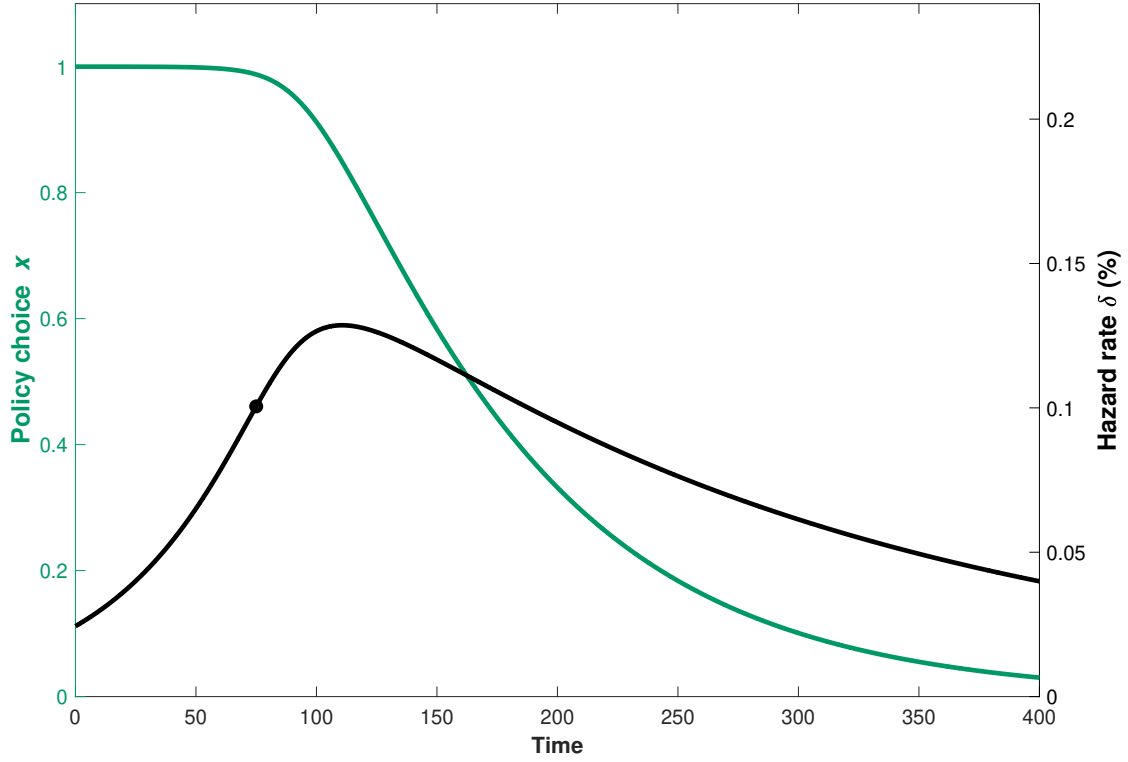


Figure 3: Evolution of the policy choice and the hazard rate along the optimal path given a lower Inada condition on safety expenditure

Derivations and code for replicating the simulation may be found in Appendix C.

²² A_0 is raised slightly in order to maintain that the value of a statistical life-year “today” (at $t = 75$) is four times per capita consumption, and the hazard rate is approximately 0.1%, despite the fact that, in this model, consumption and the hazard rate are slightly less than maximal even early in time.

A.3.2 Safety in redundancy

The constant elasticity hazard function of Sections 3.2–3.4, and its tweak just above, were chosen for clarity. We might however be interested in a better-founded story about the shape of the hazard function, in which the hazard rate is determined by the production of consumption goods and safety goods. For illustration, one relatively straightforward story would be as follows.

- Each unit of consumption (still produced as $C_t = A_t x_t$) poses some risk p of catastrophe per period in the absence of any safety measures.
- For each unit of the consumption good, if one unit of the safety good (produced as $H_t = A_t(1 - x_t)$) is allocated to preventing the production process from causing a catastrophe, this fails to prevent a catastrophe with probability $\tilde{b} < 1$. That is, one unit of H per unit of C multiplies the risk posed by each unit of C by $\tilde{b}$, from the baseline of p .
- The probability that the production of a given unit of consumption results in a catastrophe is the probability that (a) there would have been a catastrophe in the absence of any safety measures and (b) all H/C safety measures fail independently: $p\tilde{b}^{H/C}$.
- The probability that the world survives a given period is the probability that all C units of consumption, independently, do *not* generate a catastrophe: $(1 - p\tilde{b}^{H/C})^C$.

In discrete time, the story above would correspond to the hazard function

$$\delta(A_t, x_t) = 1 - \left(1 - p\tilde{b}^{\frac{1-x_t}{x_t}}\right)^{A_t x_t}, \quad \tilde{b} \in (0, 1). \quad (44)$$

The continuous-time analog to (44) is

$$\delta(A_t, x_t) = A_t x_t e^{-b \frac{1-x_t}{x_t}}, \quad b > 0 \quad (45)$$

(see Appendix B.8.1).

Since hazard function (45) lacks any sort of lower Inada condition on $1 - x$, x is fixed at 1, and δ rises, early in time while $v > 0$. After the relevant calculations, Propositions 3–5 tell us that (45) yields a Kuznets curve, with δ eventually falling quickly enough to permit survival.

Proposition 8. Long-run policy choice and risk given safety in redundancy

Given hazard function (45), the optimal path features

$$\lim_{t \rightarrow \infty} x_t t = \frac{b}{g\gamma}, \quad (46)$$

$$\lim_{t \rightarrow \infty} g_{\delta t} = -g(\gamma - 1). \quad (47)$$

Proof. See Appendix B.8.2. $\square$

Thus the decline in policy choice here is slower than in the constant elasticities case: x declines proportionally to $1/t$ rather than exponentially. This results from the fact that a model of redundancy yields a hazard rate that falls rapidly in the policy choice variable: unit decreases in $A_t x_t$, rather than merely proportional increases, generate proportional decreases to δ . In both cases, however, $x_t \rightarrow 0$. And in both cases, δ_t declines exponentially, and so quickly enough to permit survival.

Comparing (47) to the limiting expression for g_{δ} from Proposition 1, we see that, in the limit, the hazard rate declines more quickly in the redundancy-based model than in the original model. Mathematically, this follows from the fact that the extra coefficient on $g(\gamma - 1)$ in the limiting expression for g_{δ} from Proposition 1 is less than one:

$$\alpha > 0, \gamma > 1 \implies \frac{\beta - \alpha}{\beta + \gamma - 1} < 1.$$

Intuitively, this too stems from the fact that, in a redundancy-based model, smaller sacrifices in consumption (linear rather than proportional) are necessary to yield proportional decreases to the hazard rate. The planner's response to this expanded possibilities frontier comes partially in the form of slower increases in foregone consumption, as described by (46), and partially in the form of faster declines in the hazard rate, as described by (47).

A.4 Transition risk: Optimal technology growth

Without mitigation, optimality of stagnation given $\zeta = 1$

Suppose first that $\zeta = 1$ and $\alpha = -1$, so that

$$\delta_t = \bar{\delta} \frac{\dot{A}_t}{A_t}.$$

As noted in the body text, this model is precisely the Russian roulette model of Jones (2016), with $\bar{\delta}$ representing the variable there denoted π .

Jones finds in his setting that, with $\gamma > 1$, it is optimal for technology to grow only to a finite level $\hat{A}$. In our notation, this is because stagnation at some $\hat{A}$, with no risk, yields constant flow utility of $u(\hat{A})$ and a constant value of the future of $v(\hat{A}) \equiv u(\hat{A})/\rho$. It is thus optimal to halt growth at the technology level at which the future benefits of stagnating at a slightly higher A equal the costs via temporarily inducing a positive hazard rate:

$$\begin{aligned} v'(\hat{A}) &= \frac{\partial \delta}{\partial \hat{A}} \cdot v(\hat{A}) \\ \implies \frac{u'(\hat{A})}{\rho} &= \frac{\bar{\delta}}{\hat{A}} \frac{u(\hat{A})}{\rho} \end{aligned} \tag{48}$$

$$\implies \hat{A} = \left(\frac{\bar{\delta} + \gamma - 1}{\bar{\delta}} \right)^{\frac{1}{\gamma-1}}. \tag{49}$$

When $\alpha = -1$, we can derive an analytic solution for the optimal technology level (49) at which to stagnate. Though this is not possible for other values of α , it is easy to verify that, for any $\alpha \geq -\gamma$, this result does not qualitatively change. Equality (48) is then modified to

$$\begin{aligned} \frac{u'(\hat{A})}{\rho} &= \bar{\delta} \hat{A}^\alpha \frac{u(\hat{A})}{\rho} \\ \implies \hat{A}^{-(\alpha+\gamma)} &= \bar{\delta} u(\hat{A}). \end{aligned} \tag{50}$$

Given $\alpha + \gamma > 0$, the left-hand side falls strictly monotonically from 1 to 0 as $\hat{A}$ rises from 1 to ∞ . The right-hand side rises strictly monotonically from 0 to $\bar{\delta}/(\gamma - 1) > 0$ as $\hat{A}$ rises from 1 to ∞ . There is thus a unique $\hat{A} > 1$ at which (50) is satisfied: that is, at which technology growth is preferred to stagnation iff $A < \hat{A}$.

Without mitigation, no optimal stagnation given $\zeta \neq 1$

If we further generalize from $\zeta = 1$ to arbitrary ζ , however, we find that the result that stagnation is optimal when $\zeta = 1$ is knife-edge.

Let $v_t(A_{(\cdot)})$ denote the value of the future at $t \geq 0$ given technology path $A_{(\cdot)}$. As baseline, choose a technology path $A_{(\cdot)}$ satisfying A1 and A2.

If $\zeta < 1$, then at every t , and for every technology level $\bar{A} > A_t$, there is a differentiable and weakly increasing technology path $\tilde{A}_{(\cdot)}$ with $\tilde{A}_s = A_s$ for all $s \leq t$, $\tilde{A}_{\bar{t}} = \bar{A}$ for some $\bar{t} > t$, and $v_t(\tilde{A}_{(\cdot)}) > v_t(A_{(\cdot)})$.

To construct such a path, choose t and $\bar{A} > A_t$. Observe that, if $\dot{\tilde{A}}_A$ equals a constant value $\dot{\tilde{A}}$ for $A \in (A_t, \bar{A})$, the cumulative risk endured on path $\tilde{A}_{(\cdot)}$ from A_t to $\bar{A}$ equals

$$\int_{A_t}^{\bar{A}} \bar{\delta} A^\alpha \dot{\tilde{A}}^{\zeta-1} dA,$$

which $\rightarrow 0$ as $\dot{\tilde{A}} \rightarrow \infty$. With $\zeta < 1$, therefore, sufficiently rapid growth from A_t to $\bar{A}$ approximates an immediate, risk-free jump from A_t to $\bar{A}$, as in the state risk “ $\zeta = 0$ ” case.

Now let

$$\begin{aligned} \underline{t} &\equiv \min\{t : \tilde{A}_t = \bar{A}\} = \frac{\bar{A} - A_t}{\dot{\tilde{A}}}, \\ \bar{t} &\equiv \sup\{t : A_t < \bar{A}\}, \end{aligned}$$

noting that $\bar{t}$ may be infinite, and choose $\bar{A}$ and $\dot{\tilde{A}}$ so that $\dot{\tilde{A}} > \dot{A}_s$ for all $s \in [t, \underline{t}]$. This is possible for some sufficiently high $\dot{\tilde{A}}$ by the right-continuous differentiability of $A_{(\cdot)}$, and ensures that $\tilde{A}_s > A_s$ throughout this interval. Suppose that $\tilde{A}_t = \bar{A}$ for $t \in [\underline{t}, \bar{t}]$ and $\tilde{A}_t = A_t$ for $t > \bar{t}$ —i.e. that the new path halts growth at $\bar{A}$ until the old path has caught up, if ever, after which the paths are identical. Then $\tilde{A}_{(\cdot)}$ offers strictly higher consumption than $A_{(\cdot)}$ across $(t, \bar{t})$ in exchange for arbitrary little up-front risk and no subsequent increases in the hazard rate.

Incidentally, this framework makes clear that, in the absence of any costs to technological development besides transitional existential risk, with $\zeta < 1$ there is no optimal continuous technology path. An immediate jump in the technology level is always desirable, and a larger jump is always preferable to a smaller one. Furthermore, if one introduces R&D costs to the model, an optimal path will exist only if the costs are sufficiently convex in the speed of technological development. Otherwise, attempts to identify an optimal technology path will encounter the “chattering” problem: rapid alternations between slow and fast growth will be preferred to continuous growth, because they can achieve a given quantity of technological progress

over a given interval of time while contributing less to cumulative risk.

Stagnation is not optimal given $\zeta < 1$ because, due to the “upper Inada condition” on $\delta \propto \dot{A}^\zeta$ with $\zeta < 1$, sufficiently fast technological development carries arbitrarily little risk per unit of new technology. Stagnation is not optimal given $\zeta > 1$ because, since $\lim_{\dot{A} \rightarrow 0} \frac{\partial \delta}{\partial \dot{A}} = 0$ when $\delta \propto \dot{A}^\zeta$ with $\zeta > 1$, sufficiently slow technological development carries arbitrarily little risk per unit of new technology.

To see this, consider the optimal technology growth rate at t given a technology path $A_{(\cdot)}$ with $A_t = \hat{A} > 1$ and $\dot{A}_s = 0$ for $s > t$. Unlike in the $\zeta < 1$ case, there is an optimal technology growth rate to adopt at t : the rate $\dot{A}^*$ that sets the marginal expected utility benefit (via increased future consumption) of marginally increasing $\dot{A}$, per unit time that $\dot{A}$ is increased, equal to the marginal expected utility cost per unit time (via an increased hazard rate at t):

$$\begin{aligned} v'(\hat{A}) &= \bar{\delta} \hat{A}^\alpha \zeta \dot{A}^{*\zeta-1} v(\hat{A}) \\ \implies \dot{A}^* &= \left(\frac{\gamma - 1}{\bar{\delta}} \cdot \frac{\hat{A}^{-(\alpha+\gamma)}}{1 - \hat{A}^{1-\gamma}} \right)^{\frac{1}{\zeta-1}} > 0. \end{aligned}$$

Likewise, given a technology path $A_{(\cdot)}$ with $\lim_{t \rightarrow \infty} A_t = \hat{A} < \infty$, the optimal technology growth rate must satisfy the equality above in the limit. Since $A_{(\cdot)}$ cannot approach a finite upper asymptote if $\dot{A}$ is bounded above zero, no such technology path is optimal.

With mitigation, analogous results for ζ -threshold $1 + \frac{\beta}{\gamma-1}$

Throughout this section we will assume hazard function (34) with $\zeta > 0$:

$$\delta_t = \delta(A_t, \dot{A}_t, x_t) = \bar{\delta} A_t^\alpha \dot{A}_t^\zeta x_t^\beta \quad \bar{\delta} > 0, \zeta > 0, \beta > 1.$$

For simplicity we will also assume that the baseline technology path features stagnation at technology level $\hat{A}$. We will then consider the impact per unit time of an instantaneous marginal increase to the technology growth rate $\dot{A}_t$.

We will see that, in the $\zeta < 1 + \frac{\beta}{\gamma-1}$ case, as in the $\zeta < 1$ case without mitigation, there is no optimal growth rate: sufficiently fast growth is always preferable to stagnation. In the $\zeta > 1 + \frac{\beta}{\gamma-1}$ case, as in the $\zeta > 1$ case without

mitigation, growth may be “too fast”, but there is still no technology level at which it is optimal to stagnate.

However, the $\zeta = 1 + \frac{\beta}{\gamma-1}$ case is *not* closely analogous to the $\zeta = 1$ case without mitigation. Instead, for low values of $\hat{A}$ it resembles the $\zeta < 1 + \frac{\beta}{\gamma-1}$ case, with no optimal technology growth rate, and for high values of $\hat{A}$ it resembles the $\zeta < 1 + \frac{\beta}{\gamma-1}$ case, in which slow growth is preferable both to fast growth and to stagnation. Intuitively, this is because $\zeta = 1 + \frac{\beta}{\gamma-1}$ implies $\zeta > 1$. Since slow growth without mitigation is preferable to stagnation given $\zeta > 1$, and since introducing the option to mitigate risk with $x_t < 1$ does not remove the option of slow growth without mitigation, introducing the policy option cannot render stagnation optimal.

In this setting, there are two state variables: the probability of survival S and the technology level A . There are two choice variables: policy x_t and the technology growth rate $\dot{A}_t$.²³ Given $S_t = 1$, the marginal net impacts on expected utility of a marginal increase in $\dot{A}_t$, per unit time, is given by the respective derivative of the Hamiltonian expression

$$u(\hat{A}, x_t) - v_t \delta(\hat{A}, \dot{A}_t, x_t) + a_t \dot{A}_t \quad (51)$$

(adapted from Appendix B.1 below), where a is the costate variable on technology.

Under the $x_t \leq 1$ constraint, the optimal choice of x_t given $\dot{A}_t$ is given by the first order conditions $\partial \mathcal{L} / \partial x_t = 0$, $\partial \mathcal{L} / \partial \mu_t \geq 0$, $\mu_t \partial \mathcal{L} / \partial \mu_t = 0$ on the Lagrangian

$$\mathcal{L} = u(\hat{A}, x_t) - v_t \delta(\hat{A}, \dot{A}_t, x_t) + a_t \dot{A}_t + \mu(1 - x_t). \quad (52)$$

This reduces to

$$x_t = \min \left(1, \left(\bar{\delta} \beta \hat{A}^{\alpha+\gamma-1} \dot{A}_t^\zeta v(\hat{A}) \right)^{-\frac{1}{\beta+\gamma-1}} \right), \quad (53)$$

with $\mu_t > 0$ iff the second term of the above minimum—the unconstrained optimal choice of x_t —is greater than 1. (This is adapted from (61)–(62) below.)

²³It would be equivalent, and more standard but in this case more complex, to define a new choice variable ϕ_t such that the technology law of motion is $\dot{A}_t = \phi_t$.

To find the marginal net impact on expected utility of a marginal increase in $\dot{A}_t$ per unit time, given that x_t is set optimally in response, we can take the first derivative of (52) with respect to $\dot{A}_t$ and evaluate it at $x_t = (53)$. Because (51) and (52) are continuously differentiable in $\dot{A}_t$, x_t , and μ_t , by the envelope theorem we can differentiate (52) with respect to $\dot{A}_t$ and then substitute $x_t = (53)$, rather than accounting for the impact of changing $\dot{A}_t$ on the choice of x_t by substituting (53) into (51) and differentiating the result with respect to $\dot{A}_t$.

Finally, given technology level $A_t = \hat{A}$ and permanent stagnation after t , the value of the costate variables at t are straightforward. The value of [saving] civilization at t is $v(\hat{A})$, and the value of a marginal increase in the technology level is the value of an equal marginal increase in consumption at all future periods:

$$\begin{aligned} v_t = v(\hat{A}) &= \frac{1}{\rho} \cdot \frac{\hat{A}^{1-\gamma} - 1}{1 - \gamma}, \\ a_t = v'(\hat{A}) &= \frac{\hat{A}^{-\gamma}}{\rho}. \end{aligned}$$

The marginal net impact on expected utility of a marginal increase in $\dot{A}_t$ per unit time is therefore

$$\begin{aligned} d(\dot{A}_t) &\equiv \frac{\hat{A}^{-\gamma}}{\rho} - v(\hat{A})\bar{\delta}\hat{A}^\alpha\zeta\dot{A}_t^{\zeta-1}x_t^\beta \\ &= \frac{\hat{A}^{-\gamma}}{\rho} - v(\hat{A})\bar{\delta}\hat{A}^\alpha\zeta\dot{A}_t^{\zeta-1}, & \dot{A}_t < \underline{\dot{A}}_t; \\ &= \frac{\hat{A}^{-\gamma}}{\rho} - \zeta\left(\bar{\delta}^{1-\gamma}v(\hat{A})^{1-\gamma}\hat{A}^{(\beta-\alpha)(\gamma-1)}\beta^\beta\right)^{-\frac{1}{\beta+\gamma-1}}\dot{A}_t^{\zeta\frac{\gamma-1}{\beta+\gamma-1}-1}, & \dot{A}_t \geq \underline{\dot{A}}_t, \end{aligned} \tag{54}$$

where

$$\underline{\dot{A}}_t \equiv (\bar{\delta}\beta\hat{A}^{\alpha+\gamma-1}v(\hat{A}))^{-\frac{1}{\zeta}}$$

is the maximum growth rate at which it is optimal to set $x_t = 1$, and $v(\hat{A})$ is as defined above.

If $\zeta < 1 + \frac{\beta}{\gamma-1}$, then the exponent on $\dot{A}_t$ in (54) is negative for $\dot{A}_t \geq \underline{\dot{A}}_t$, so

$$\lim_{\dot{A}_t \rightarrow \infty} d(\dot{A}_t) = \hat{A}^{-\gamma}/\rho > 0.$$

As in the $\zeta < 1$ case without policy, this guarantees that sufficiently fast technology growth is always preferable to stagnation.

If $\zeta > 1 + \frac{\beta}{\gamma-1}$, then the exponent on $\dot{A}_t$ in (54) is always positive. There is thus a unique and positive value of $\dot{A}_t$ that sets $d(\dot{A}_t) = 0$, and this is the optimal choice of $\dot{A}_t$. Sufficiently slow technology growth is always preferable to stagnation.

If $\zeta = 1 + \frac{\beta}{\gamma-1}$, then the exponent on $\dot{A}_t$ in (54) is positive for $\dot{A}_t < \underline{\dot{A}}_t$ and zero for $\dot{A}_t \geq \underline{\dot{A}}_t$. So if $d(\underline{\dot{A}}_t) > 0$, there is no optimal growth rate: from the $A_t = \hat{A}$ margin, it is desirable, albeit perhaps briefly, to have technology grow as quickly as possible. If $d(\underline{\dot{A}}_t) < 0$, there is a unique value of $\dot{A}_t$ that sets $d(\dot{A}_t) = 0$, it lies in $(0, \underline{\dot{A}}_t)$, and it is optimal.

Technically, if $d(\underline{\dot{A}}_t) = 0$, then any $\dot{A}_t \geq \underline{\dot{A}}_t$ is optimal at $A_t = \hat{A}$; but once $A_t > \hat{A}$, we will have $d(\underline{\dot{A}}_t) < 0$, and a unique optimal growth rate which is positive but finite.

B Proofs

B.1 Existence and uniqueness of optimal policy

Necessary and sufficient conditions

The dynamic optimization problems analyzed in Sections 3–4 all feature one choice variable x and one state variable S . Expected flow utility at t is $S_t u(A_t, x_t)$ for a twice continuously differentiable function $u(\cdot)$, strictly concave in x , with a lower Inada condition on x . The law of motion for S is given by $-S_t \delta(A_t, \dot{A}_t, x_t)$ for a twice continuously differentiable function $\delta(\cdot)$. A and $\dot{A}$ are independent of x , so operate simply as functions of t .

Letting v denote the costate variable on S , the current value Lagrangian corresponding to the problem is then

$$\mathcal{L}(S_t, x_t, v_t, \mu_t, t) = S_t u(x_t, t) - v_t S_t \delta(x_t, t) + \mu_t (1 - x_t) \quad (55)$$

(abusing notation slightly by reusing $u(\cdot)$ and $\delta(\cdot)$ as functions of time), where μ_t represents the the Lagrange multiplier on x_t . We impose the $x_t \leq 1$ constraint but not the $x_t \geq 0$ constraint because the latter can never bind, by the lower Inada condition on $u(\cdot)$.

(55) satisfies the Mangasarian concavity condition that $\mathcal{L}(\cdot)$ is everywhere concave in S and x . So, applying Caputo (2005), Theorems 14.3-4 and

Lemma 14.1,²⁴ given continuous paths of $x \in [0, 1]$ and $S \in [0, 1]$ with $S_0 = 1$ and $\dot{S}_t = -S_t \delta(x_t, t)$, we have that the x, S path is optimal if—and, given piecewise continuity of x and S , only if—for some piecewise differentiable path of v and some piecewise continuous path of $\mu \geq 0$, at all t the following first-order conditions are satisfied

$$\frac{\partial \mathcal{L}}{\partial x_t}(S_t, x_t, v_t, \mu_t, t) = 0, \quad (56)$$

$$\frac{\partial \mathcal{L}}{\partial \mu_t}(S_t, x_t, v_t, \mu_t, t) \geq 0, \quad (57)$$

$$\mu_t \frac{\partial \mathcal{L}}{\partial \mu_t}(S_t, x_t, v_t, \mu_t, t) = 0 \quad (58)$$

as well as the transversality condition that

$$\lim_{t \rightarrow \infty} e^{-\rho t} v_t = \lim_{t \rightarrow \infty} e^{-\rho t} v_t S_t = 0. \quad (59)$$

Furthermore, given optimal paths of x and S and corresponding paths of v and μ , v will satisfy

$$\begin{aligned} \dot{v}_t &= \rho v_t - \frac{\partial \mathcal{L}}{\partial S_t} = \rho v_t - u(x_t, t) - v_t \dot{S}_t \\ &= (\rho + \delta(x_t, t))v_t - u(x_t, t) \end{aligned} \quad (60)$$

except at any discontinuity points of x , at which v will have different right and left derivatives.

Interpreting the transversality condition

Given a continuous v path, only the paths of x and μ defined by

$$x_t = \begin{cases} 1, & \frac{\partial u}{\partial x}(1, t) - \frac{\partial \delta}{\partial x}(1, t)v_t \geq 0; \\ x_t : \frac{\partial u}{\partial x}(x_t, t) - \frac{\partial \delta}{\partial x}(x_t, t)v_t = 0, & \text{otherwise} \end{cases} \quad (61)$$

$$\mu_t = \frac{\partial u}{\partial x_t}(x_t, t) - \frac{\partial \delta}{\partial x_t}(x_t, t)v_t \quad (62)$$

satisfy (56)–(58) for all t . Any such x path is well-defined, by the continuous differentiability of $u(\cdot)$ and $\delta(\cdot)$ in x and the fact that $u(\cdot)$ and $\delta(\cdot)$ strictly increase in x . Any such x path is also right-continuous in time, by

²⁴Caputo (2005) uses the more general present value notation. Because the control problem at hand is exponentially discounted, we here use the simpler current value notation.

- the twice continuous differentiability of $u(\cdot)$ and $\delta(\cdot)$ (expressed as functions of x , A , and perhaps, in the case of $\delta(\cdot)$, $\dot{A}$);
- the right-continuous differentiability of $A(\cdot)$ in time;
- the right-continuous differentiability of $\dot{A}(\cdot)$ assumed in conjunction with the hazard functions considered in Section 4;

and the implicit function theorem. Any such μ path is then also right-continuous in time by the composition of continuous functions. To show there exists an optimal path, and that only one such path is piecewise continuous, it will now suffice to show that there is a unique v path for which (59)–(60) are satisfied given the corresponding x path (61) and its implied S path, and that the corresponding x path is piecewise continuous (in fact it is right-continuous).

The solution to differential equation (60) is

$$v_t = e^{\int_0^t (\rho + \delta_s) ds} \left(v_0 - \int_0^t e^{-\int_0^s (\rho + \delta_q) dq} u(x_s, s) ds \right) \quad (63)$$

$$\implies v_0 = \int_0^t e^{-\rho s} S_s u(x_s, s) ds + e^{-\rho t} S_t v_t. \quad (64)$$

Since (64) is continuous in t (by the boundedness of $u(\cdot)$ and the continuous evolution of S) and holds for all t , v satisfies (59)–(60) iff

$$v_0 = \int_0^\infty e^{-\rho t} S_t u(x_t, t) dt. \quad (65)$$

That is, the value of decreasing the probability of a catastrophe at time 0 (as of time 0) must equal the expected utility of the future (as of time 0, given survival up to time 0).

Given (61), v_t determines x_t for all t , and given (60), v_t and x_t determine $\dot{v}_t$ for all t . For a given v_0 , therefore, there is a unique path of v —and thus of x , and thus of S —compatible with (60)–(61). We will now show that there is at least one value of v_0 for which (65) is satisfied, given the corresponding x and S paths. For such a v_0 , the corresponding variable paths will by construction satisfy (56)–(59), and thus constitute an optimum.

Existence

Let $v(v_0)$ and $x(v_0)$ denote the unique paths of v and x compatible with (60)–(61) for which $v_0(v_0) = v_0$. By (63), $\lim_{v_0 \rightarrow -\infty} v_t(v_0) = -\infty$ for all $t \geq 0$. By (61), therefore, for every $t \geq 0$, there is a $\tilde{v}_0$ such that $x_t(v_0) = 1$ for all $v_0 < \tilde{v}_0$. Let $s \geq 0$ denote a time at which $A_s \geq 1$, and choose $\tilde{v}_0$ low enough that $\tilde{v}_s < 0$ and thus $x_s(\tilde{v}_0) = 1$. By (60), because $u(1, s) \geq 0$, $\dot{v}_t < 0$. We thus have $\tilde{v}_t < 0$, and thus $x_t = 1$, for all $t \geq s$.

Now observe that if $v_0 < \tilde{v}_0$, $v_t(v_0) < v_t(\tilde{v}_0)$ for all t . Otherwise, by the continuity of v with respect to time, there would be a t with $v_t(v_0) = v_t(\tilde{v}_0)$, and integrating (60), with (61) substituted for x_t , would allow us to identify $v_0 = \tilde{v}_0$. Thus, if $v_0 < \tilde{v}_0$, $x_t(v_0) \geq x_t(\tilde{v}_0)$ for all $t \geq 0$. It follows that, for some sufficiently low $\underline{v}_0$, the right-hand side of (65) exceeds the left-hand side.

For every optimization problem under consideration, there is some $\bar{U}$ by which feasible values of the right-hand side of (65) are upper-bounded. So, for $\bar{v}_0 > \bar{U}$, the left-hand side of (65) exceeds the right-hand side.

By (61), the implicit function theorem gives us that x_t is continuous (indeed, continuously differentiable except at one point) in v_t for all t . (60) then implies that $\dot{v}_t$ is continuous in v_t for all t , and thus that $v_t(v_0)$, then $x_t(v_0)$, and then ultimately the right-hand side of (65) are continuous in v_0 for all t . It follows from the intermediate value theorem that there exists a $v_0 \in (\underline{v}_0, \bar{v}_0)$ for which (65) holds.

Uniqueness

The uniqueness result of Caputo (2005), Theorem 14.4 (cited above) does not immediately apply here, because the Lagrangian is linear, not strictly concave, in the state variable S . Fortunately, this can easily be remedied by defining the state variable to be e.g. S^2 without affecting any conditions necessary for the other results.

Uniqueness (among piecewise continuous x paths) also follows immediately from the observations that a path is optimal iff v_0 attains its maximum feasible value and that, given (56)–(59), v_0 determines a unique path for every variable.

B.2 Long-run g_v and proof of Proposition 2

Long-run constancy of g_v for all γ

From (60), because v is the costate variable on S , it must follow the law of motion

$$\begin{aligned} \dot{v}_t &= (\rho + \delta_t)v_t - u(C_t) \\ \implies g_{vt} &= \rho + \delta(A_t, x_t) - \frac{u(A_t x_t)}{v_t}. \end{aligned} \quad (66)$$

Let

$$\tilde{\beta} \equiv \beta + \gamma - 1.$$

From (15), once x_t is interior we have

$$x_t = A_t^{-\frac{\alpha+\gamma-1}{\beta}} (\bar{\delta}\beta v_t)^{-\frac{1}{\beta}}. \quad (67)$$

Substituting (67) into (66) yields

$$g_{vt} = g_v(v_t, t) \equiv \begin{cases} \rho + K A_t^{\frac{(\beta-\alpha)(1-\gamma)}{\beta}} v_t^{-\frac{\beta}{\beta}} + \frac{1}{1-\gamma} v_t^{-1}, & \gamma \neq 1; \\ \rho + \log(A_t^{-\frac{\beta-\alpha}{\beta}} (\bar{\delta}\beta v_t)^{-\frac{1}{\beta}}) v_t^{-1}, & \gamma = 1, \end{cases} \quad (68)$$

where

$$K \equiv \bar{\delta}^{-\frac{1-\gamma}{\beta}} (\beta^{-\frac{\beta}{\beta}} - \frac{1}{1-\gamma} \beta^{-\frac{1-\gamma}{\beta}}).$$

If $\gamma > 1$, recalling that v_t monotonically increases and that $A_t \rightarrow \infty$, the central term of (68) vanishes. Also, in this case, v is upper-bounded, so it approaches an upper bound v^* by the monotone convergence theorem. So $\lim_{t \rightarrow \infty} g_{vt}$ is defined, with

$$\lim_{t \rightarrow \infty} g_{vt} = \rho + \frac{1}{v^*(1-\gamma)}. \quad (69)$$

This limit cannot be positive, because v is upper-bounded, and it cannot be negative, because v increases with time. So $\lim_{t \rightarrow \infty} g_{vt} = 0$, and $v^* = \frac{1}{\rho(\gamma-1)}$.

If $\gamma < 1$, then $K < 0$, and the central term of (68) grows in magnitude without bound, fixing v . v must therefore also grow without bound, or else g_{vt} is eventually negative.

Now observe that

$$\begin{aligned} \dot{g}_{vt} &= K A_t^{\frac{(\beta-\alpha)(1-\gamma)}{\tilde{\beta}}} v_t^{-\frac{\beta}{\tilde{\beta}}} \left(\frac{(\beta-\alpha)(1-\gamma)}{\tilde{\beta}} g - \frac{\beta}{\tilde{\beta}} g_{vt} \right) - \frac{1/v_t}{1-\gamma} g_{vt} \\ &= \left(g_{vt} - \rho - \frac{1/v_t}{1-\gamma} \right) \left(\frac{(\beta-\alpha)(1-\gamma)}{\tilde{\beta}} g - \frac{\beta}{\tilde{\beta}} g_{vt} \right) - \frac{1/v_t}{1-\gamma} g_{vt} \\ &= -\frac{\beta}{\tilde{\beta}} g_{vt}^2 + \left(\frac{(\beta-\alpha)(1-\gamma)}{\tilde{\beta}} g + \frac{\beta}{\tilde{\beta}} \rho + \frac{1}{\tilde{\beta} v_t} \right) g_{vt} - \left(\rho + \frac{1/v_t}{1-\gamma} \right) \frac{(\beta-\alpha)(1-\gamma)}{\tilde{\beta}} g. \end{aligned}$$

This differential equation has two steady states, both positive. Since $1/v_t \rightarrow 0$, the quadratic formula tells us that these steady states approach ρ and $g(\beta-\alpha)(1-\gamma)/\beta$, with the former attractive and the latter repulsive. By (21), ρ is higher, and is ruled out as a steady state by the transversality condition (59). Then because the limits

$$\begin{aligned} \lim_{t \rightarrow \infty} \dot{g}_v(g_v, t) &> 0 \quad \forall g_v \in \left(\frac{(\beta-\alpha)(1-\gamma)}{\beta} g, \rho \right), \\ \lim_{t \rightarrow \infty} \dot{g}_v(g_v, t) &< 0 \quad \forall g_v < \frac{(\beta-\alpha)(1-\gamma)}{\beta} g \end{aligned}$$

are defined and continuous in g_v , we must have

$$\lim_{t \rightarrow \infty} g_{vt} = \frac{(\beta-\alpha)(1-\gamma)}{\beta} g. \quad (70)$$

Otherwise we would have $g_v \rightarrow -\infty$, ruled out by the monotonicity of v , or $g_v \rightarrow \rho$, ruled out above.

The $\gamma = 1$ case is analogous to the $\gamma > 1$ case. Differentiating (68) with respect to time yields $\dot{g}_{vt}$ strictly and continuously increasing in g_{vt} from $-\infty$ at $v_t = 0$ to ρ at $v_t = \infty$. There is thus a unique, positive, and repulsive “time-dependent steady state” value of g_v (i.e. g_v for which $\dot{g}_v(g_v, t) = 0$) which declines to zero as $t \rightarrow \infty$. The limits

$$\begin{aligned} \lim_{t \rightarrow \infty} \dot{g}_v(g_v, t) &> 0 \quad \forall g_v > 0, \\ \lim_{t \rightarrow \infty} \dot{g}_v(g_v, t) &< 0 \quad \forall g_v < 0 \end{aligned}$$

are defined and continuous in g_v , and we must have

$$\lim_{t \rightarrow \infty} g_{vt} = 0$$

to avoid $g_v \rightarrow -\infty$ or $g_v \rightarrow \infty$.

Proof of Proposition 2

With the limiting behavior of g_v pinned down, the asymptotic behavior of the other variables follows straightforwardly. Substituting (70) for g_{vt} into expression (16) for g_{xt} (and observing that the expression captures all $\gamma \leq 1$) produces

$$\lim_{t \rightarrow \infty} g_{xt} = -\frac{\alpha}{\beta}g,$$

and adding αg then produces the limit of $g_{Ax} = g_C$:

$$\lim_{t \rightarrow \infty} g_{Ct} = \frac{\beta - \alpha}{\beta}g. \quad (71)$$

For the hazard rate, rearrange (68) to get

$$v_t = \frac{u(C_t)}{\rho + \delta_t - g_{vt}}, \quad (72)$$

and substitute (72) into (25) to get

$$\delta_t = \begin{cases} \frac{\rho + \delta_t - g_{vt}}{\beta} \frac{1 - \gamma}{1 - C_t^{\gamma-1}}, & \gamma < 1; \\ \frac{\rho + \delta_t - g_{vt}}{\beta \log(C_t)}, & \gamma = 1. \end{cases}$$

Solving for δ_t ,

$$\delta_t = \begin{cases} \frac{(\rho - g_{vt})(1 - \gamma)}{\beta(1 - C_t^{\gamma-1}) - 1 + \gamma}, & \gamma < 1; \\ \frac{\rho - g_{vt}}{\beta \log(C_t) - 1}, & \gamma = 1. \end{cases}$$

In the $\gamma < 1$ case, the limit of g_v (70) and $C \rightarrow \infty$ from (71) imply

$$\lim_{t \rightarrow \infty} \delta_t = \frac{(\rho - (\beta - \alpha)(1 - \gamma)g/\beta)(1 - \gamma)}{\beta + \gamma - 1}.$$

In the $\gamma = 1$ case, substitute 0 for g_{vt} and observe that, by (71),

$$\lim_{t \rightarrow \infty} \frac{C_t}{e^{\frac{\beta-\alpha}{\beta}gt}} = \underline{C}$$

for some $\underline{C} > 0$, so that

$$\begin{aligned} \lim_{t \rightarrow \infty} \delta_t t &= \lim_{t \rightarrow \infty} \frac{\rho - g_{vt}}{\beta (\log(C_t / e^{\frac{\beta-\alpha}{\beta}gt}) + \log(e^{\frac{\beta-\alpha}{\beta}gt})) / t - 1/t} \\ &= \lim_{t \rightarrow \infty} \frac{\rho}{\beta \log(\underline{C}) / t + (\beta - \alpha)g - 1/t} \\ &= \frac{\rho}{(\beta - \alpha)g}. \end{aligned}$$

B.3 Proof of Proposition 3

Suppose that $R^* \leq 1$, and, by contradiction, that we do not have $C^* = \infty$.

By the failure of $C^* = \infty$, there is an increasing and unbounded sequence of times, $t_n \rightarrow \infty$, such that $C_{t_n} \leq \bar{C} \forall n \geq 1$.

Consider the sequence of consumption levels $n\bar{C} \forall n \geq 1$. Since $n\bar{C} \rightarrow \infty$, by $R^* \leq 1$ we have

$$\lim_{n \rightarrow \infty} R(n\bar{C}) = \lim_{n \rightarrow \infty} \lim_{A \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A, \frac{n\bar{C}}{A} \right) \frac{(n\bar{C})^\gamma}{A\rho(\gamma-1)} \leq 1. \quad (73)$$

By D5, $\frac{\partial \delta}{\partial x}(A, x)$ weakly increases in x for any A . So

$$R(C_{t_n}) \leq R(n\bar{C}) \left(\frac{C_{t_n}}{n\bar{C}} \right)^\gamma \leq R(n\bar{C}) n^{-\gamma} \quad \forall n, \quad (74)$$

where the first inequality follows from the fact that $n\bar{C} \geq C_{t_n}$ for each n , and the second follows from $\bar{C} \geq C_{t_n}$ for each n . By (73), $R(n\bar{C})n^{-\gamma} < 1$ for sufficiently large n , so by (74) and A4, there exists an $\underline{n}$ such that

$$\frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{C_{t_n}}{A_{t_n}} \right) \frac{C_{t_n}^\gamma}{A_{t_n}\rho(\gamma-1)} < 1 \quad \forall n > \underline{n}.$$

Since v_t cannot exceed $\frac{1}{\rho(\gamma-1)}$,

$$\frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{C_{t_n}}{A_{t_n}} \right) v_{t_n} < A_{t_n} C_{t_n}^{-\gamma} \quad \forall n > \underline{n}.$$

This is compatible with optimality only if $x_{t_n} = 1$. But this is impossible for sufficiently large n , since $C_{t_n} = A_{t_n}x_{t_n} \leq \bar{C}$ and $\lim_{n \rightarrow \infty} A_{t_n} = \infty$.

Suppose that $R^* > 1$ and, by contradiction, that $C^* = \infty$. Then there is some $\underline{C}$ such that $R(\underline{C}) > 1$:

$$\lim_{A \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A, \frac{C}{A} \right) \frac{\underline{C}^\gamma}{A \rho(\gamma - 1)} > 1.$$

So there is an $\underline{A}$ such that

$$\frac{\partial \delta}{\partial x} \left(A, \frac{C}{A} \right) \frac{1}{\rho(\gamma - 1)} > A \underline{C}^{-\gamma} \quad (75)$$

for all $A \geq \underline{A}$. Furthermore, because the left-hand side weakly increases in $\underline{C}$ by D5 and the right-hand side strictly decreases in $\underline{C}$, (75) holds for all $A \geq \underline{A}$ and $C \geq \underline{C}$. By A4, and the supposition that $C^* = \infty$, there is a $\underline{t}$ such that

$$\frac{\partial \delta}{\partial x} \left(A_t, \frac{C_t}{A_t} \right) \frac{1}{\rho(\gamma - 1)} > A_t C_t^{-\gamma} \quad \forall t \geq \underline{t}. \quad (76)$$

Finally, optimality requires

$$\begin{aligned} A_t^{1-\gamma} x_t^{-\gamma} &\geq \frac{\partial \delta}{\partial x_t} (A_t, x_t) v_t \quad \forall t \\ \implies (A_t x_t)^{1-\gamma} / v_t &\geq \frac{\partial \delta}{\partial x_t} (A_t, x_t) x_t \geq \delta(A_t, x_t), \end{aligned}$$

with the final inequality holding because, by D5, $\frac{\partial \delta}{\partial x} x \geq \delta$. Given $C^* = \infty$, since v_t is upper-bounded, it follows that $\delta_t \rightarrow 0$. With $\delta_t \rightarrow 0$ and $C_t \rightarrow \infty$, v_t approaches its upper bound of $\frac{1}{\rho(\gamma-1)}$.

It therefore follows from (76) that, for sufficiently large t ,

$$\frac{\partial \delta}{\partial x} \left(A_t, \frac{C_t}{A_t} \right) v_t > A_t C_t^{-\gamma}.$$

This is incompatible with optimality. Thus, if $R^* > 1$, it is impossible that $C^* = \infty$.

B.4 Proof of Proposition 4a

It is optimal to set $x_t = 1$ as long as, at $x = 1$, the marginal flow disutility of decreasing x weakly exceeds the marginal expected utility of doing so via decreasing the hazard rate:

$$A_t^{1-\gamma} \geq \frac{\partial \delta}{\partial x}(A_t, 1) v_t. \quad (77)$$

It is optimal to set $x_t < 1$ as long as (77) fails, maintaining

$$A_t^{1-\gamma} x_t^{-\gamma} = \frac{\partial \delta}{\partial x}(A_t, x_t) v_t \quad (78)$$

$$\implies x_t = A_t^{\frac{1-\gamma}{\gamma}} \left(\frac{\partial \delta}{\partial x}(A_t, x_t) v_t \right)^{-\frac{1}{\gamma}}. \quad (79)$$

The uniqueness of the optimal path is shown in Appendix B.1.

Proof that $\lim_{t \rightarrow -\infty} x_t = 1$

We will show that there exists a time $\underline{t}$ such that $v_{\underline{t}} \leq 0$. It then follows immediately that $x_t = 1$ for $t \leq \underline{t}$.

Let

$$T \equiv A^{-1}((\gamma - 1)^{\frac{1}{1-\gamma}})$$

denote the time at which $A_T = (\gamma - 1)^{\frac{1}{1-\gamma}}$, and at which therefore $u(A_T) = -1$. If $v_T \leq 0$, the result follows immediately. Let us therefore assume that $v_T > 0$.

For $t < T$,

$$\begin{aligned} v_t &= \int_t^\infty e^{-\rho(s-t) - \int_t^s \delta_q dq} u(C_s) ds \\ &= \int_t^T e^{-\rho(s-t) - \int_t^s \delta_q dq} u(C_s) ds + e^{-\rho(T-t) - \int_t^T \delta_q dq} v_T. \end{aligned} \quad (80)$$

Since $u(C_s) \leq u(A_s) \leq -1$ for $s \leq T$, the first term of (80) is negative—indeed, an integral over s of values which are negative for all s . The integral is shrunk in magnitude when, for all s , $u(C_s)$ is replaced with -1 and the

discount factor $e^{-\rho(s-t)-\int_t^s \delta_q dq}$ replaced with its minimum value across the range, namely the discount factor at T . So

$$\begin{aligned} v_t &< (t - T + v_T) e^{-\rho(T-t)-\int_t^T \delta_q dq} \\ \implies v_{T-v_T} &< 0. \end{aligned}$$

This proof admittedly “takes the model too literally”, in assuming that technology growth has always been exponential and that therefore life was not worth living before some point in the past. Still, the dynamic it bluntly illustrates should not be controversial. When $\gamma > 1$, proportional sacrifices in consumption—decreases to x —carry greater utility costs the lower the baseline consumption level is. Early in time, the discounted value of civilization v and the baseline consumption level A were both low, so large sacrifices for safety would not have been optimal.

Proof that $\lim_{t \rightarrow \infty} x_t = 0$ if η_A is bounded above $1 - \gamma$

Generalizing (79), whether or not the $x_t \leq 1$ constraint binds we have

$$x_t \leq A_t^{\frac{1-\gamma}{\gamma}} \left(\frac{\partial \delta}{\partial x}(A_t, x_t) v_t \right)^{-\frac{1}{\gamma}}. \quad (81)$$

We will show that if $\eta_A(\cdot)$ is bounded above $1 - \gamma$, the right-hand side has an upper bound which falls to 0 as (by A4) $A_t \rightarrow \infty$.

Because by D1 δ is positive, by D2 and D5 we have $\frac{\partial \delta}{\partial x}(A_t, x_t) \geq \delta(A_t, x_t)$. The right-hand side is thus bounded above by

$$A_t^{\frac{1-\gamma}{\gamma}} (\delta(A_t, x_t) v_t)^{-\frac{1}{\gamma}}. \quad (82)$$

Fixing x and v , the elasticity of this upper bound with respect to A is $(1 - \gamma - \eta_A(A, x))/\gamma$. Since this is here bounded below 0, (82) tends to 0 as $A \rightarrow \infty$. Finally, v_t is positive for all $t \geq 0$, because by A1 and A2 $A_t > 1$ for all $t \geq 0$ (rendering $v_t > 0$ feasible with $x = 1$ permanently), and v_t does not fall because sufficient precautions on new technology—e.g. banning its use—allow the consumption path to be maintained without increasing risk, by D4. Therefore, if $\eta_A(\cdot)$ is bounded above $1 - \gamma$, maintaining optimality condition (81) as $A_t \rightarrow \infty$ requires $x_t \rightarrow 0$.

B.5 Proof of Proposition 5

If $\lim_{k \downarrow 1} \tilde{R}(k) < 1$, there is a $\bar{k} > 1$ such that

$$\lim_{t \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_t, \frac{t^{\frac{\bar{k}}{\gamma-1}}}{A_t} \right) \frac{t^{\frac{\bar{k}\gamma}{\gamma-1}}}{A_t \rho(\gamma-1)} < 1. \quad (83)$$

Choose $k \in (1, \bar{k})$. Suppose that $\nexists \underline{t} : C_t > t^{\frac{k}{\gamma-1}} \quad \forall t > \underline{t}$. Then there is an increasing and unbounded sequence of times, $\{t_n\} \rightarrow \infty$, such that

$$C_{t_n} \leq t_n^{\frac{k}{\gamma-1}} \quad \forall n \geq 1. \quad (84)$$

Observe that

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{t_n^{\frac{k}{\gamma-1}}}{A_{t_n}} \right) \frac{t_n^{\frac{k\gamma}{\gamma-1}}}{A_{t_n} \rho(\gamma-1)} \\ & \leq \lim_{t \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_t, \frac{t^{\frac{\bar{k}}{\gamma-1}}}{A_t} \right) \frac{t^{\frac{\bar{k}\gamma}{\gamma-1}}}{A_t \rho(\gamma-1)} \cdot t^{-\frac{\bar{k}-k}{\gamma-1}\gamma} = 0, \end{aligned} \quad (85)$$

where the inequality follows from the fact that, by D5, $\frac{\partial \delta}{\partial x}(A, x)$ weakly increases in x , and the limit before the $t^{-\frac{\bar{k}-k}{\gamma-1}\gamma}$ term is less than 1 by (83).

By (84), (85), and the fact that $v_t < \frac{1}{\rho(\gamma-1)}$ for all t , there is an $\underline{n}$ such that, for all $n \geq \underline{n}$,

$$\frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{C_{t_n}}{A_{t_n}} \right) v_{t_n} < A_{t_n} C_{t_n}^{-\gamma}.$$

This is compatible with optimality only if $x_{t_n} = A_{t_n} x_{t_n} = 1$. But this is impossible for sufficiently large n , by (31) and (84).

So for some $k > 1$,

$$\exists \underline{t} : C_t > t^{\frac{k}{\gamma-1}} \quad \forall t > \underline{t}. \quad (86)$$

So (86) holds for $k = 1$ as well.

Given (86) for some $k > 1$, we have, for some $\underline{t}$ and some $\underline{k} \in (1, k)$, that for all $t > \underline{t}$

$$\begin{aligned}
 & (A_t x_t)^{1-\gamma} < t^{-k} \\
 \implies & \frac{\partial \delta}{\partial x}(A_t, x_t) x_t v_t < t^{-k} \\
 \implies & \delta_t v_t < t^{-k} \\
 \implies & \delta_t < t^{-\underline{k}}.
 \end{aligned} \tag{87}$$

The first implication follows from the fact that $A_t^{1-\gamma} x_t^{-\gamma} \geq \frac{\partial \delta}{\partial x}(A_t, x_t) v_t$ whether or not x is interior. The second follows from the fact that $\delta < \frac{\partial \delta}{\partial x} x$ by D1 and D5. The third follows from the fact that v_t is eventually positive and does not fall to zero.

δ_t is uniformly bounded from 0 to $\underline{t}$ by $\max_{A \in [A_0, A_{\underline{t}}]} \delta(A, 1)$, which exists and is finite by the continuity of $\delta(\cdot)$ (D3). It follows from this and from (87) that $S_\infty > 0$.

If $\lim_{k \uparrow 1} \tilde{R}(k) > 1$, there is a $k < 1$ and an $\underline{s}$ such that

$$\frac{\partial \delta}{\partial x} \left(A_t, \frac{t^{\frac{k}{\gamma-1}}}{A_t} \right) \frac{t^{\frac{k\gamma}{\gamma-1}}}{A_t \rho(\gamma-1)} > 1 \quad \forall t > \underline{s}. \tag{88}$$

Suppose by contradiction that $\nexists \underline{t} : C_t < t^{\frac{1}{\gamma-1}} \quad \forall t > \underline{t}$. Then there is an increasing and unbounded sequence of times, $\{t_n\} \rightarrow \infty$, such that

$$C_{t_n} \geq t_n^{\frac{1}{\gamma-1}} \quad \forall n \geq 1. \tag{89}$$

Observe that

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{t_n^{\frac{1}{\gamma-1}}}{A_{t_n}} \right) \frac{t_n^{\frac{\gamma}{\gamma-1}}}{A_{t_n} \rho(\gamma-1)} \\
 & \geq \lim_{t \rightarrow \infty} \frac{\partial \delta}{\partial x} \left(A_t, \frac{t^{\frac{k}{\gamma-1}}}{A_t} \right) \frac{t^{\frac{k\gamma}{\gamma-1}}}{A_t \rho(\gamma-1)} \cdot t^{\frac{1-k}{\gamma-1} \gamma} = \infty,
 \end{aligned} \tag{90}$$

where the inequality follows from the fact that, by D5, $\frac{\partial \delta}{\partial x}(A, x)$ weakly increases in x , and the limit before the $t^{\frac{1-k}{\gamma-1} \gamma}$ term is greater than 1 by (88).

By (89), (90), and the fact that $v_t \not\rightarrow 0$, there is an n such that

$$\frac{\partial \delta}{\partial x} \left(A_{t_n}, \frac{C_{t_n}}{A_{t_n}} \right) v_{t_n} > A_{t_n} C_{t_n}^{-\gamma}.$$

This is incompatible with optimality. So

$$\exists \underline{t} : C_t < t^{\frac{1}{\gamma-1}} \quad \forall t > \underline{t}. \quad (91)$$

By (91) and (31), $x_t \rightarrow 0$. So there exists a $\bar{t} \geq \underline{t}$ such that, for all $t > \bar{t}$, the choice of x is interior

$$\frac{\partial \delta}{\partial x} (A_t, x_t) v_t = A_t^{1-\gamma} x_t^{-\gamma}$$

and so, by (91),

$$\frac{\partial \delta}{\partial x} (A_t, x_t) x_t v_t = C_t^{1-\gamma} > 1/t.$$

Since $\eta_x \equiv \frac{\partial \delta}{\partial x} \frac{x}{\delta}$,

$$\eta_x(A_t, x_t) \delta(A_t, x_t) v_t > 1/t \quad \forall t \geq \bar{t}.$$

Recall that an interior choice of x_t implies that $v_t > 0$, that v is upper-bounded by $\frac{1}{\rho(\gamma-1)}$, and that $\delta_t > 0$ by D1. So $\eta_x > 0 \quad \forall t \geq \bar{t}$. So if η_x is upper-bounded by $\bar{\eta}_x$,

$$\delta(A_t, x_t) > \frac{\rho(\gamma-1)}{\bar{\eta}_x} \cdot \frac{1}{t} \quad \forall t \geq \bar{t}.$$

So $S_\infty = 0$.

B.6 Proof of Proposition 6

Choose an admissible technology path $A_{(\cdot)}$ and hazard function $\delta(\cdot)$.

Choose $\underline{A}$, $\dot{\tilde{A}}$ with $\dot{\tilde{A}} > \dot{A}_{\underline{A}}$. Define $\tilde{x}_A[\epsilon]$ as $\tilde{x}_A$ given acceleration $\tilde{A}_{(\cdot)}[\epsilon]$, etc. v_t is weakly increasing and continuous (indeed differentiable; see Appendix B.1) in t . Since A_t is continuous, increasing, and invertible in t , v_A

is continuous and weakly increasing in A . $v_{\underline{A}+\epsilon}$ is therefore continuous and weakly increasing in ϵ .

From technology level $\underline{A} + \epsilon$ onward, the technology paths, and thus the paths of both consumption and the hazard rate, are identical under $A_{(\cdot)}$ and $\tilde{A}_{(\cdot)}$. So for any ϵ (including 0), $\tilde{v}_{\underline{A}+\epsilon}[\epsilon] = v_{\underline{A}+\epsilon}$. From this, the fact that $\tilde{v}_A[\epsilon]$ is weakly increasing in A , and the fact that $\tilde{v}_{\underline{A}}[\epsilon] \geq v_{\underline{A}}$ for all ϵ , we have that for all ϵ

$$\tilde{v}_A[\epsilon] \in [\tilde{v}_{\underline{A}}, \tilde{v}_{\underline{A}+\epsilon}] \subseteq [v_{\underline{A}}, v_{\underline{A}+\epsilon}] \quad \forall A \in [\underline{A}, \underline{A} + \epsilon]. \quad (92)$$

Then by the continuity of $v_{\underline{A}+\epsilon}$ in ϵ , for any ϵ_1 there is an $\bar{\epsilon}$ such that $|v_{\underline{A}+\epsilon} - v_{\underline{A}}| < \epsilon_1 \quad \forall \epsilon < \bar{\epsilon}$.

Adapting (61),

$$\tilde{x}_A[\epsilon] = \min \left(1, x : \frac{\partial \delta}{\partial x}(A, x) A^{\gamma-1} x^\gamma = \frac{1}{\tilde{v}_A[\epsilon]} \right). \quad (93)$$

By (93) and A2, $\tilde{v}_A[\epsilon] \geq v_{\underline{A}} > 0$ for all $\epsilon \geq 0, A \in [\underline{A}, \underline{A} + \epsilon]$. By D3, the implicit function theorem, and the continuity of $\min(\cdot)$, $\tilde{x}_A[\epsilon]$ is continuous in $\tilde{v}_A[\epsilon]$. So by (92) and the sentence following it, for any ϵ_2 there is an $\bar{\epsilon}$ such that, for all $\epsilon < \bar{\epsilon}$,

$$\left| \tilde{x}_A[\epsilon] - \min \left(1, x : \frac{\partial \delta}{\partial x}(A, x) A^{\gamma-1} x^\gamma = \frac{1}{v_{\underline{A}}} \right) \right| < \epsilon_2 \quad \forall A \in [\underline{A}, \underline{A} + \epsilon].$$

Again by D3, the implicit function theorem, and the continuity of $\min(\cdot)$, the second term in the absolute value is continuous in A . So for any ϵ_3 there is an $\bar{\epsilon}$ such that, for all $\epsilon < \bar{\epsilon}$,

$$\begin{aligned} \left| \tilde{x}_A[\epsilon] - \min \left(1, x : \frac{\partial \delta}{\partial x}(\underline{A}, x) \underline{A}^{\gamma-1} x^\gamma = \frac{1}{v_{\underline{A}}} \right) \right| \\ = \left| \tilde{x}_A[\epsilon] - x_{\underline{A}} \right| < \epsilon_3 \quad \forall A \in [\underline{A}, \underline{A} + \epsilon]. \end{aligned}$$

With this uniform convergence, since

$$\tilde{X}[\epsilon] - X = \int_{\underline{A}}^{\underline{A}+\epsilon} \delta(A, \tilde{x}_A[\epsilon]) \dot{A}^{-1} dA - \int_{\underline{A}}^{\underline{A}+\epsilon} \delta(A, x_A) \dot{A}_A^{-1} dA,$$

since $\delta(\cdot)$ is continuous in both arguments, since x_A is continuous in A , and since $\dot{A}_A^{-1}$ is right-continuous in time and thus (by the continuity and

monotonicity of $A_{(\cdot)}$ in A ,

$$\begin{aligned}\Delta_{\underline{A}, \dot{A}} &\equiv \lim_{\epsilon \rightarrow 0} \frac{\tilde{X}[\epsilon] - X}{\epsilon} = \delta(\underline{A}, x_{\underline{A}}) \dot{A}^{-1} - \delta(\underline{A}, x_{\underline{A}}) \dot{A}_{\underline{A}}^{-1} \\ &= \delta_{\underline{A}}(\dot{A}^{-1} - \dot{A}_{\underline{A}}^{-1}).\end{aligned}$$

This proves (a).

Let $\tilde{A}_{(\cdot)}$ be an acceleration to $A_{(\cdot)}$ from $\underline{A}$ to $\bar{A}$. By the definition of an acceleration and the definition of cumulative risk,

$$\tilde{X} = X + \int_{\underline{A}}^{\bar{A}} \left(\delta(A, \tilde{x}_A) \dot{A}_A^{-1} - \delta(A, x_A) \dot{A}_A^{-1} \right) dA. \quad (94)$$

For all $A \in [\underline{A}, \bar{A}]$, we have $\tilde{v}_A \geq v_A$, and thus, by (93) (dropping the “[ϵ]” arguments) and D5, $\tilde{x}_A \leq x_A$. D1, D2, and D5 imply that $\delta(\cdot)$ weakly increases in x , so $\delta(A, \tilde{x}_A) \leq \delta(A, x_A)$. So

$$\delta(A, \tilde{x}_A) \dot{A}_A^{-1} - \delta(A, x_A) \dot{A}_A^{-1} \leq \Delta_{A, \dot{A}_A} \quad \forall A \in [\underline{A}, \bar{A}].$$

This proves (b).

If $\bar{A} < \infty$, the integral of (94) finite. So given a technology path $A_{(\cdot)}$ for which $X = \infty$ and an acceleration to $\bar{A} < \infty$, $\tilde{X} = \infty$. This proves the first part of (c).

To prove the second part of (c), it will suffice to find a hazard function $\delta(\cdot)$ and technology path $A_{(\cdot)}$ for which $X = \infty$ and a pair of accelerations $\tilde{A}_{(\cdot)}$ to $\bar{A} = \infty$, for one of which $\tilde{X}$ is finite and for the other of which $\tilde{X}$ is infinite. We have already encountered both.

For a case of the former, consider the hazard function $\delta(A_t, x_t) = A_t x_t$, discussed following Proposition 5. As discussed there, cumulative risk given optimal policy is then infinite for any technology path eventually bounded above zero.

For a case of the latter, consider hazard function (5)— $\delta(A_t, x_t) = \bar{\delta} A_t^\alpha x_t^\beta$ —with baseline technology path $A_t = (t - 1)^k$ ($t \geq 0$) and acceleration $\tilde{A}_t = (t - 1)^{\tilde{k}}$ ($t \geq 0$), where

$$k \leq \frac{\beta + \gamma - 1}{(\alpha - \beta)(\gamma - 1)} < \tilde{k}.$$

To verify that this is an acceleration, $A_t = (t - 1)^k \implies t = 1 + A_t^{\frac{1}{k}}$, so $\dot{A}_t = k(t - 1)^{k-1} \implies \dot{A}_A = kA^{\frac{k-1}{k}}$, which increases in k given $A > 1$ (which holds for $t > 0$).

As explained in Section 2.2.2 (footnote 16), here $X = \infty$ and $\tilde{X} < \infty$.

B.7 Proof of Proposition 7

The proof is similar to the proof of Proposition 6a (Appendix B.6). As there, $v_{\underline{A}+\epsilon}$ is continuous in ϵ and $\tilde{v}_{\underline{A}+\epsilon}[\epsilon] = v_{\underline{A}+\epsilon}$ for all ϵ . In this setting, however, we cannot assume that $\tilde{v}_A[\epsilon]$ weakly increases in A or that $\tilde{v}_A[\epsilon] \geq v_A$ for all ϵ . We will therefore use a different strategy to uniformly bound $\tilde{v}_A[\epsilon]$, for $A \in [\underline{A}, \underline{A} + \epsilon]$, in an interval whose maximum and minimum converge to $v_{\underline{A}}$ as $\epsilon \rightarrow 0$.

Let $\underline{t}$ denote the time at which $A_t = \underline{A}$. An acceleration $\tilde{A}_{(\cdot)}$, featuring technology growth rate $\dot{\tilde{A}} > \dot{A}_{\underline{A}}$ until technology level $\underline{A} + \epsilon$, features technology growth at rate $\dot{\tilde{A}}$ across times

$$(\underline{t}, \underline{t} + \epsilon / \dot{\tilde{A}}).$$

More generally, the acceleration path reaches technology level $A \in [\underline{A}, \underline{A} + \epsilon]$ at time

$$\tilde{t}(A) \equiv \underline{t} + (A - \underline{A}) / \dot{\tilde{A}}.$$

$\tilde{v}_A[\epsilon]$ is the maximum value of survival $\tilde{v}_{\tilde{t}(A)}$, across feasible policy paths, achievable at $\tilde{t}(A)$ given technology path $\tilde{A}_{(\cdot)}[\epsilon]$. It can thus be lower-bounded by one such achievable value of survival, such as that achieved given $x_t = 1$ for $t \in [\tilde{t}(A), \underline{t} + \epsilon / \dot{\tilde{A}}]$. Since $\tilde{A}_t > 1$ throughout this interval, this lower bound is in turn strictly greater than the value of survival at $\tilde{t}(A)$ given *no* flow utility enjoyed throughout the interval.

Remembering that $\tilde{v}_{\underline{A}+\epsilon}[\epsilon] = v_{\underline{A}+\epsilon} > 0$ for any ϵ , we thus have

$$\begin{aligned} \tilde{v}_A[\epsilon] &\geq \int_{\tilde{t}(A)}^{\underline{t} + \epsilon / \dot{\tilde{A}}} e^{-\rho(t - \tilde{t}(A))} e^{-\int_{\tilde{t}(A)}^t \bar{\delta} \tilde{A}_s^\alpha \dot{\tilde{A}}^\zeta ds} u(\tilde{A}_t) dt \\ &\quad + e^{-\rho(\underline{t} + \epsilon / \dot{\tilde{A}} - \tilde{t}(A))} e^{-\int_{\tilde{t}(A)}^{\underline{t} + \epsilon / \dot{\tilde{A}}} \bar{\delta} \tilde{A}_s^\alpha \dot{\tilde{A}}^\zeta ds} v_{\underline{A}+\epsilon} \\ &> v_A[\epsilon] \equiv e^{-\rho(\underline{t} + \epsilon / \dot{\tilde{A}} - \tilde{t}(A))} e^{-\int_{\tilde{t}(A)}^{\underline{t} + \epsilon / \dot{\tilde{A}}} \bar{\delta} \tilde{A}_s^\alpha \dot{\tilde{A}}^\zeta ds} v_{\underline{A}+\epsilon}. \end{aligned} \tag{95}$$

Because $\tilde{t}(A)$ increases in A , $\underline{v}_A[\epsilon]$ increases in A , so $\underline{v}_A[\epsilon] \geq \underline{v}_{\underline{A}}[\epsilon]$ for all $A \in [\underline{A}, \underline{A} + \epsilon]$.

$\tilde{v}_A[\epsilon]$ can be upper-bounded by the (infeasible) value of survival achieved at $\tilde{t}(A)$ given that, at $t \in [\tilde{t}(A), \underline{t} + \epsilon/\dot{A}]$, flow utility equals its supremum of $1/(\gamma - 1)$ and the hazard rate equals 0:

$$\begin{aligned} \tilde{v}_A[\epsilon] &< \frac{1}{\gamma - 1} \int_{\tilde{t}(A)}^{\underline{t} + \epsilon/\dot{A}} e^{-\rho(t - \tilde{t}(A))} dt + e^{-\rho(\underline{t} + \epsilon/\dot{A} - \tilde{t}(A))} \underline{v}_{\underline{A} + \epsilon} \\ &< \bar{v}_A[\epsilon] \equiv \frac{1}{\gamma - 1} \int_{\tilde{t}(A)}^{\underline{t} + \epsilon/\dot{A}} e^{-\rho(t - \tilde{t}(A))} dt + \underline{v}_{\underline{A} + \epsilon}. \end{aligned} \quad (96)$$

Because $\tilde{t}(A)$ increases in A , $\underline{v}_A[\epsilon]$ decreases in A , so $\underline{v}_A[\epsilon] \geq \underline{v}_{\underline{A} + \epsilon}[\epsilon]$ for all $A \in [\underline{A}, \underline{A} + \epsilon]$.

From (95), (96), the continuity of $\underline{v}_{\underline{A} + \epsilon}$ in ϵ , and the fact that $\tilde{v}_{\underline{A} + \epsilon}[\epsilon] = \underline{v}_{\underline{A} + \epsilon}$ for all ϵ ,

$$\lim_{\epsilon \rightarrow 0} \underline{v}_{\underline{A}}[\epsilon] = \lim_{\epsilon \rightarrow 0} \bar{v}_{\underline{A} + \epsilon}[\epsilon] = \underline{v}_{\underline{A}}.$$

The proof then proceeds along the lines of the proof of Proposition 6 after (92), with

$$\tilde{x}_A[\epsilon] = \min \left(1, \left(\bar{\delta} \beta A^{\alpha + \gamma - 1} \dot{A} \tilde{v}_A[\epsilon] \right)^{-\frac{1}{\beta + \gamma - 1}} \right)$$

in place of (93), ultimately yielding

$$\Delta_{\underline{A}, \dot{A}} = \delta(\underline{A}, \dot{A}, \tilde{x}_{\underline{A}}) \dot{A}^{-1} - \delta(\underline{A}, \dot{A}_{\underline{A}}, x_{\underline{A}}) \dot{A}_{\underline{A}}^{-1}, \quad (97)$$

where $\tilde{x}_{\underline{A}}$ is given by (36), at $A = \underline{A}$, with $\dot{A}$ in place of $\dot{A}_{\underline{A}}$.

If $\underline{A} \geq A^*$, (97) reduces to

$$\left(\bar{\delta}^{1-\gamma} \beta^\beta \underline{A}^{(\beta-\alpha)(\gamma-1)} v_{\underline{A}}^\beta \right)^{-\frac{1}{\beta+\gamma-1}} \left(\dot{A}^{\zeta \frac{\gamma-1}{\beta+\gamma-1}-1} - \dot{A}_{\underline{A}}^{\zeta \frac{\gamma-1}{\beta+\gamma-1}-1} \right).$$

Since $\dot{A} > \dot{A}_{\underline{A}}$, this is negative if $\zeta < 1 + \frac{\beta}{\gamma-1}$, zero if $\zeta = 1 + \frac{\beta}{\gamma-1}$, and positive if $\zeta > 1 + \frac{\beta}{\gamma-1}$.

If $\underline{A} < A^*$, so that $x_{\underline{A}} = 1$, and $\dot{A}$ is small enough to maintain $\tilde{x}_{\underline{A}} = 1$, then (97) reduces to

$$\bar{\delta} \underline{A}^\alpha (\dot{A}^{\zeta-1} - \dot{A}_{\underline{A}}^{\zeta-1}).$$

Since $\dot{A} > \dot{A}_{\underline{A}}$, this is negative if $\zeta < 1$, zero if $\zeta = 1$, and positive if $\zeta > 1$.

B.8 Safety in redundancy

B.8.1 From discrete to continuous

Suppose a unit of production carries a constant flow probability $\bar{\delta}$ of triggering an existential catastrophe, so that, in the absence of any safeguards, the probability that it does not trigger a catastrophe after s units of time is $e^{-\bar{\delta}s}$. To be consistent with the discrete-time specification that the probability that it triggers a catastrophe after 1 unit of time equals p , we have $1 - e^{-\bar{\delta}} = p$ and thus $\bar{\delta} = -\log(1 - p)$.

With $\frac{1-x_t}{x_t}$ units of safeguards maintained around t , since each unit multiplies the probability of a catastrophic failure per unit time by a factor $\tilde{b} \in (0, 1)$, we have that the probability that a catastrophe is avoided until $t + s$ equals $e^{-\bar{\delta}\tilde{b}\frac{1-x}{x}s}$.

The probability that $A_t x_t$ equally-safeguarded units of production all avoid catastrophe until $t + s$ is thus

$$\left(e^{-\bar{\delta}\tilde{b}\frac{1-x_t}{x_t}s}\right)^{A_t x_t} = e^{-\bar{\delta}\tilde{b}\frac{1-x_t}{x_t}A_t x_t s}. \quad (98)$$

So the probability of a catastrophe by s given locally constant A, x equals 1-(98), and the hazard rate—the probability of catastrophe per unit time—at time t precisely is

$$\delta_t \equiv \lim_{s \rightarrow 0} (1 - e^{-\bar{\delta}\tilde{b}\frac{1-x_t}{x_t}A_t x_t s})/s = \bar{\delta}A_t x_t \tilde{b}^{\frac{1-x_t}{x_t}}.$$

Letting $b \equiv -\log(\tilde{b}) > 0$ yields

$$\delta_t = \bar{\delta}A_t x_t e^{-b\frac{1-x_t}{x_t}}.$$

B.8.2 Proof of Proposition 8

By Appendix B.1, there is a unique optimal path. By the reasoning following (10), the optimal choice of x is 1 until the (unique) time at which

$$\frac{\partial u}{\partial x_t}(A_t, x_t) = \frac{\partial \delta}{\partial x_t}(A_t, x_t) v_t \quad (99)$$

at $x_t = 1$, after which the optimal choice of x_t is interior and maintains equality (99).

Differentiating the utility function and hazard function (45), we have

$$\begin{aligned} A_t^{1-\gamma} x_t^{-\gamma} &= \bar{\delta} A_t e^{-b \frac{1-x_t}{x_t}} \left(1 + \frac{b}{x_t}\right) v_t \\ \implies \frac{1}{v_t} &= \bar{\delta} A_t^\gamma e^{-b \frac{1-x_t}{x_t}} \left(x_t^\gamma + b x_t^{\gamma-1}\right). \end{aligned} \quad (100)$$

Because v_t increases monotonically and is upper-bounded, it is asymptotically positive and constant, by the monotone convergence theorem.

We must have $C_t \rightarrow \infty$. If we do not, then there is a unbounded sequence of times t_n and a consumption level $\bar{C}$ such that

$$x_{t_n} \leq \bar{C}/A_{t_n} \quad \forall n. \quad (101)$$

Substituting (101) into (100), and recalling that $A_{t_n} \rightarrow \infty$, this would imply that the right-hand side of (100) tends to 0 across $\{t_n\}$, and thus that it is not asymptotically positive.

From (100),

$$\frac{1}{v_t} = \delta_t C_t^{\gamma-1} (1 + b/x_t).$$

Since $C_t^{\gamma-1} \rightarrow \infty$, x_t cannot be negative, and $1/v_t \not\rightarrow \infty$, it follows that $\delta_t \rightarrow 0$.

Since $C_t \rightarrow \infty$ and $\delta_t \rightarrow 0$, $v_t \rightarrow \bar{v}$.

Divide both sides of (100) by $\bar{\delta} A_0^\gamma$, and take the log and then the limit. With

$$\kappa \equiv \log \left(A_0^{-\gamma} \frac{1}{\rho(\gamma-1)\bar{\delta}} \right),$$

we have

$$\begin{aligned} \lim_{t \rightarrow \infty} \left[g\gamma t - b \frac{1-x_t}{x_t} + \log \left(x_t^\gamma + b x_t^{\gamma-1} \right) \right] &= \kappa \\ \implies \lim_{t \rightarrow \infty} \frac{x_t}{1-x_t} t &= \lim_{t \rightarrow \infty} \frac{b}{g\gamma - \kappa/t + \log \left(x_t^\gamma + b x_t^{\gamma-1} \right) / t}. \end{aligned}$$

Other than $g\gamma$, the terms in the denominator on the right-hand side must converge to 0. This would be avoided only if there were an unbounded sequence of times t_n across which x_{t_n} grew at least exponentially with time,

which is impossible, or shrank at least exponentially with time, which would send the right-hand side of (100) to zero. So

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{x_t}{1 - x_t} t &= \frac{b}{g\gamma} \\ \implies \lim_{t \rightarrow \infty} x_t t &= \lim_{t \rightarrow \infty} (1 - x_t) \frac{b}{g\gamma} = \frac{b}{g\gamma} \\ \implies \lim_{t \rightarrow \infty} x_t \frac{g\gamma}{b} t &= 1, \end{aligned}$$

since $x_t \rightarrow 0$. It then follows from the hazard function that, in the limit, δ falls to 0 at exponential rate $-g(\gamma - 1) < 0$.

C Transition dynamics for simulations

For simulating the transition dynamics, it is helpful to find $\dot{x}_t$ and $\dot{\delta}_t$ as functions of t and x_t in the regime where x is interior.

Hazard function (5), used throughout Sections 3.2–3.4 and used to simulate Figures 1 and 2, is the special case of hazard function (43), used to simulate Figure 3, with $\epsilon = 1$. The calculations below therefore apply to all simulations.

FOC:

$$\begin{aligned} \frac{\partial u}{\partial x_t}(A_t, x_t) &= \frac{\partial \delta}{\partial x_t}(A_t, x_t) v_t \\ \implies A_t^{1-\gamma} x_t^{-\gamma} &= \bar{\delta} A_t^\alpha x_t^{\beta-2} \left((\beta - 1)(1 - (1 - x_t)^\epsilon) + \epsilon x_t (1 - x_t)^{\epsilon-1} \right) v_t. \end{aligned}$$

Rearranging and differentiating gives

$$v_t = \frac{1}{\bar{\delta}} \frac{A_t^{1-\gamma-\alpha} x_t^{2-\gamma-\beta}}{(\beta - 1)(1 - (1 - x_t)^\epsilon) + \epsilon x_t (1 - x_t)^{\epsilon-1}} \quad (102)$$

$$\begin{aligned} \implies \dot{v}_t &= v_t \left((1 - \gamma - \alpha)g + (2 - \gamma - \beta) \frac{\dot{x}_t}{x_t} \right. \\ &\quad \left. - \epsilon \frac{\beta - (\epsilon + \beta - 1)x_t}{(\beta - 1)(1 - x_t)^{1-\epsilon} + 1 - \beta + (\epsilon + \beta - 1)x_t} \frac{\dot{x}_t}{1 - x_t} \right). \end{aligned} \quad (103)$$

From the first-order condition with respect to the state variable S_t ,

$$\begin{aligned}\dot{v}_t &= v_t(\rho + \delta_t) - u(c_t) \\ &= v_t\left(\rho + \bar{\delta}A_t^\alpha x_t^{\beta-1}(1 - (1 - x_t)^\epsilon)\right) - \frac{(A_t x_t)^{1-\gamma} - 1}{1 - \gamma}.\end{aligned}\quad (104)$$

Substituting (102) into (103) and (104), setting the results equal, and solving for $\dot{x}_t$ yields

$$\begin{aligned}\dot{x}_t &= x_t((\beta - 1)(1 - x_t)^{1-\epsilon} + 1 - \beta + (\epsilon + \beta - 1)x_t)(1 - x_t) \\ &\quad \left((2 - \gamma - \beta)((\beta - 1)(1 - x_t)^{1-\epsilon} + 1 - \beta \right. \\ &\quad \left. + (\epsilon + \beta - 1)x_t)(1 - x_t) - \epsilon(\beta - (\epsilon + \beta - 1)x_t)x_t\right)^{-1} \\ &\quad \left(\rho + \bar{\delta}A_t^\alpha x_t^{\beta-1}(1 - (1 - x_t)^\epsilon) - g(1 - \alpha - \gamma) - \right. \\ &\quad \left. \frac{(A_t x_t)^{1-\gamma} - 1}{1 - \gamma} \bar{\delta}A_t^{\alpha+\gamma-1} x_t^{\beta+\gamma-2}((\beta - 1)(1 - (1 - x_t)^\epsilon) + \epsilon x_t(1 - x_t)^{\epsilon-1})\right).\end{aligned}\quad (105)$$

Differentiating the hazard function (43) with respect to t yields

$$\dot{\delta}_t = \bar{\delta}A_t^\alpha x_t^\beta \frac{1 - (1 - x_t)^\epsilon}{x_t} \left(\alpha g + (\beta - 1)\frac{\dot{x}_t}{x_t} + \epsilon \frac{(1 - x_t)^\epsilon}{1 - (1 - x_t)^\epsilon} \frac{\dot{x}_t}{1 - x_t}\right). \quad (106)$$

Scripts for replicating Figures 1, 2, and 3 using (105) and (106), and the estimate of S_∞ following Figure 1, are provided here: https://philiptrammell.com/static/ERAG_code.zip.